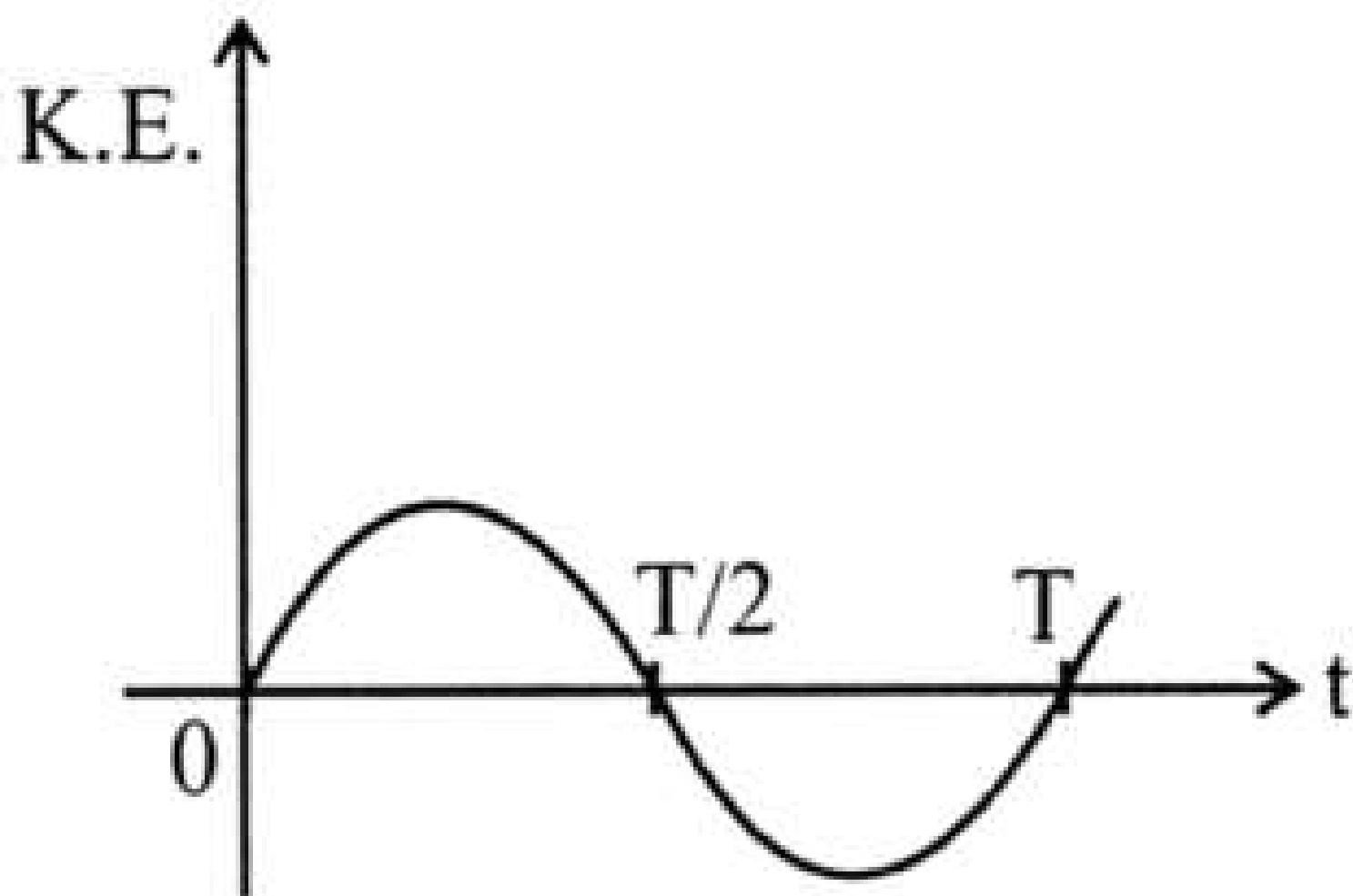


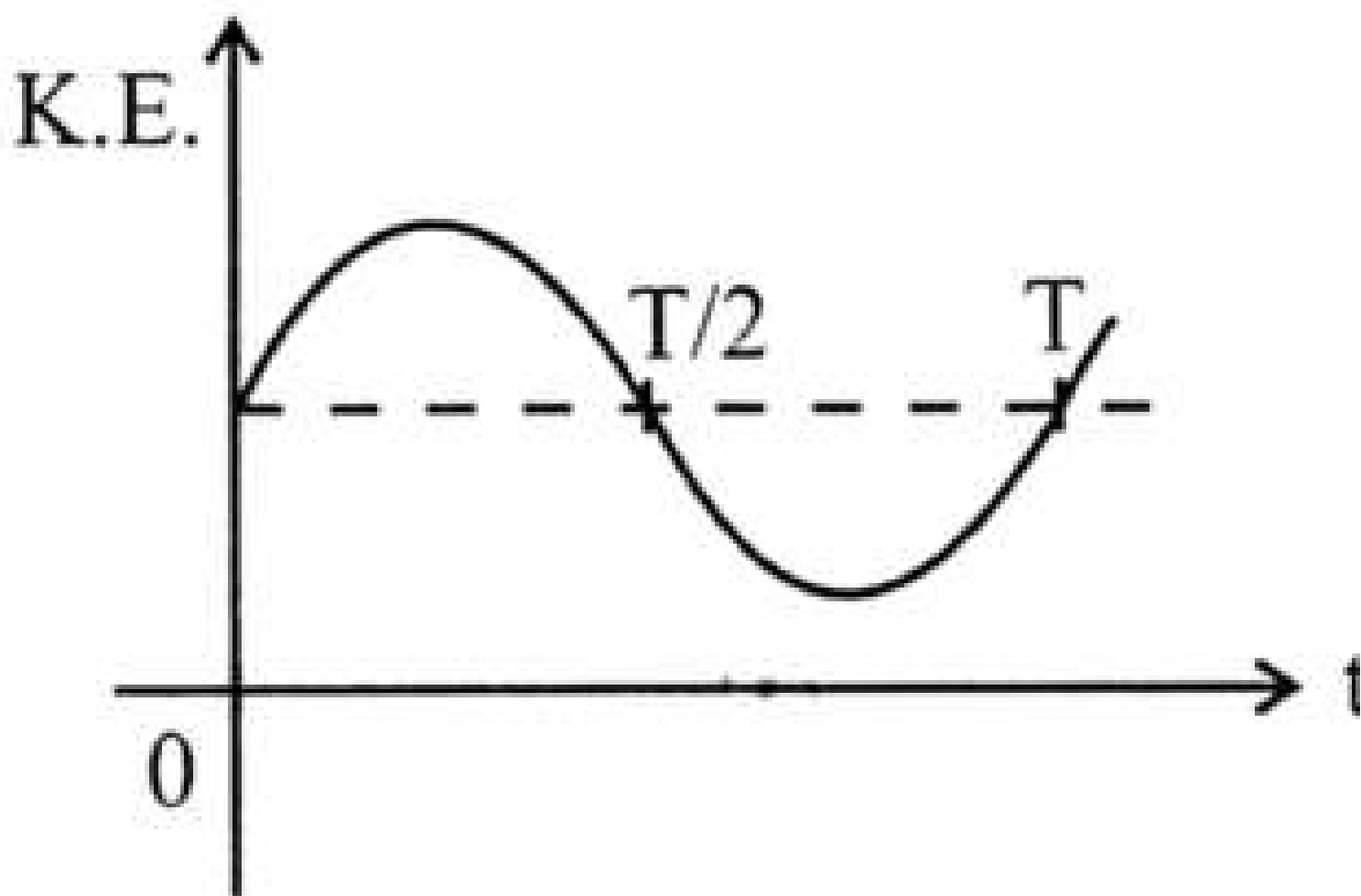
- 1) 2.0 m/s
3) 14.1 m/s

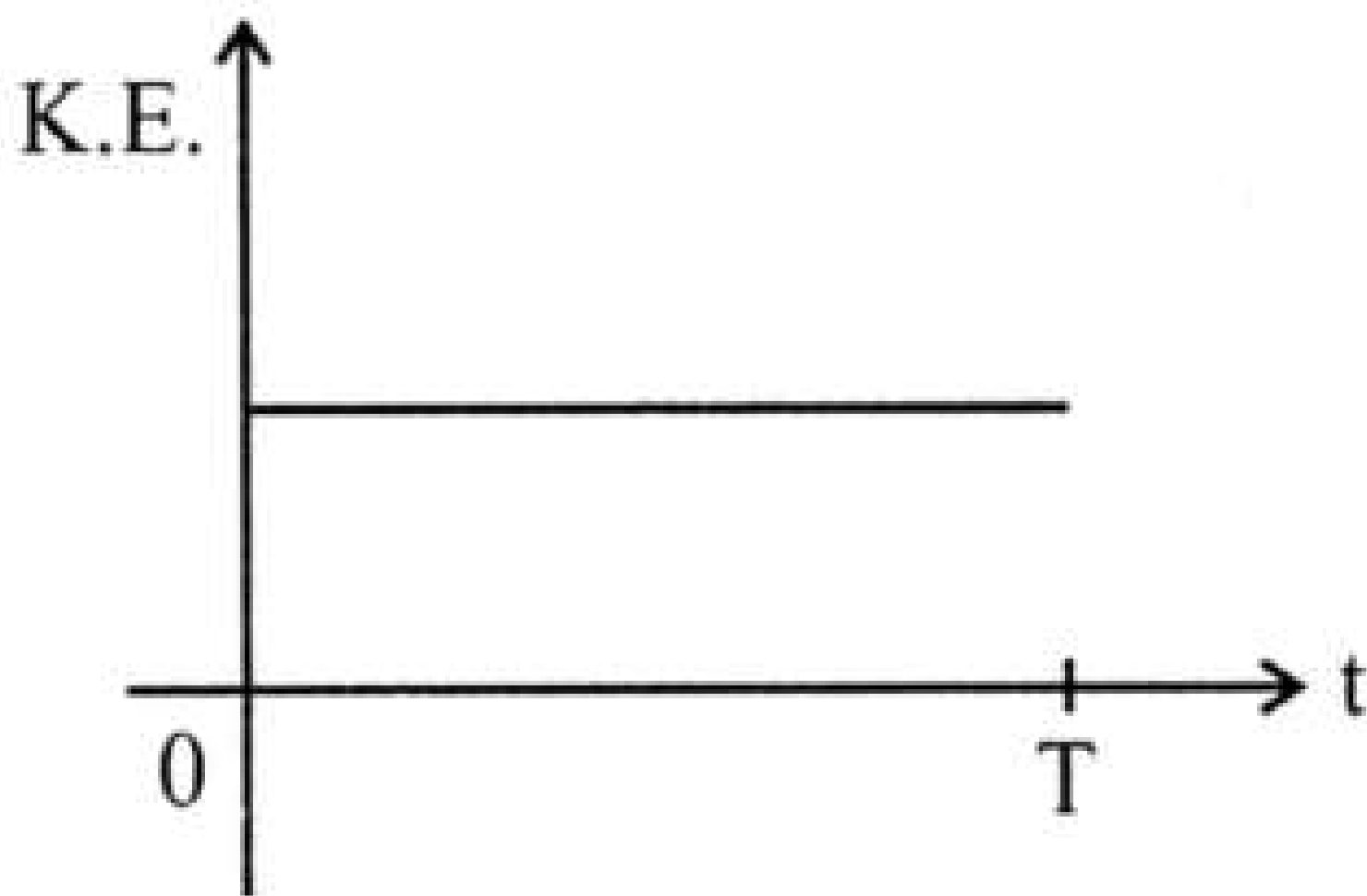
Sol. (4):

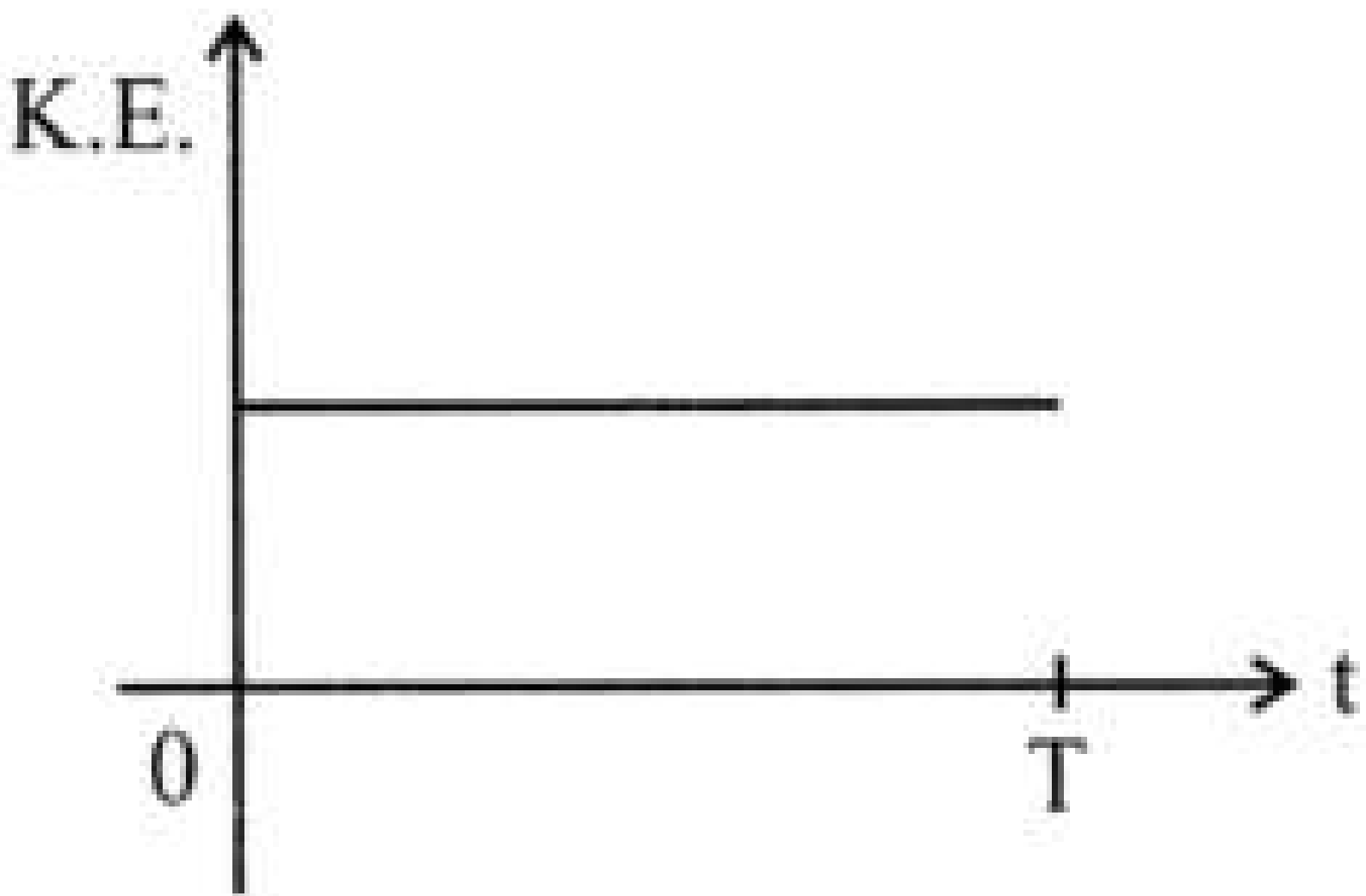
Magnetic field

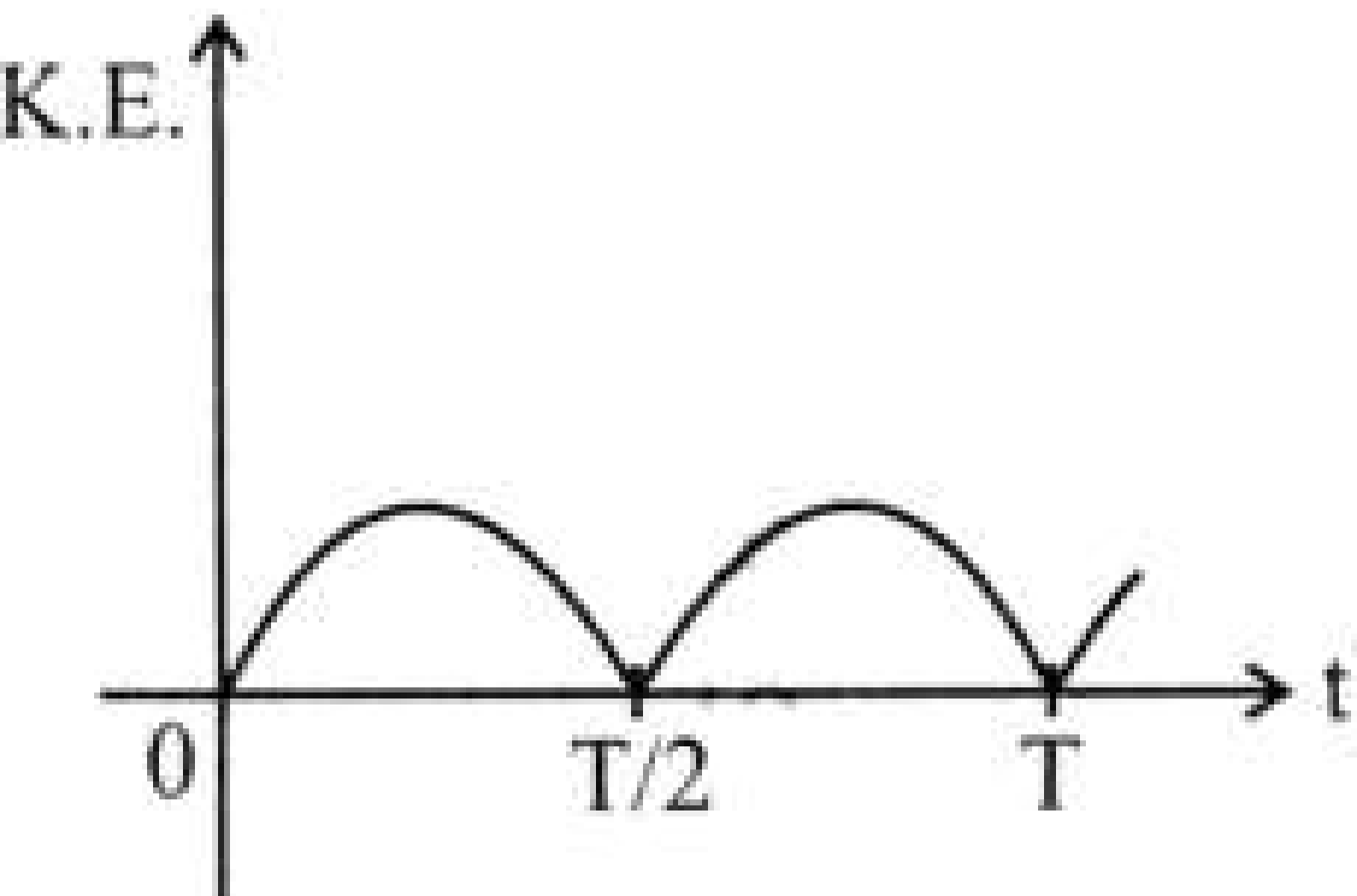
5. A 100 turn coil of radius 5 cm has a current of 2.0 A at its centre. Find the magnetic field at the centre of the coil, and the magnetic flux through the coil, and

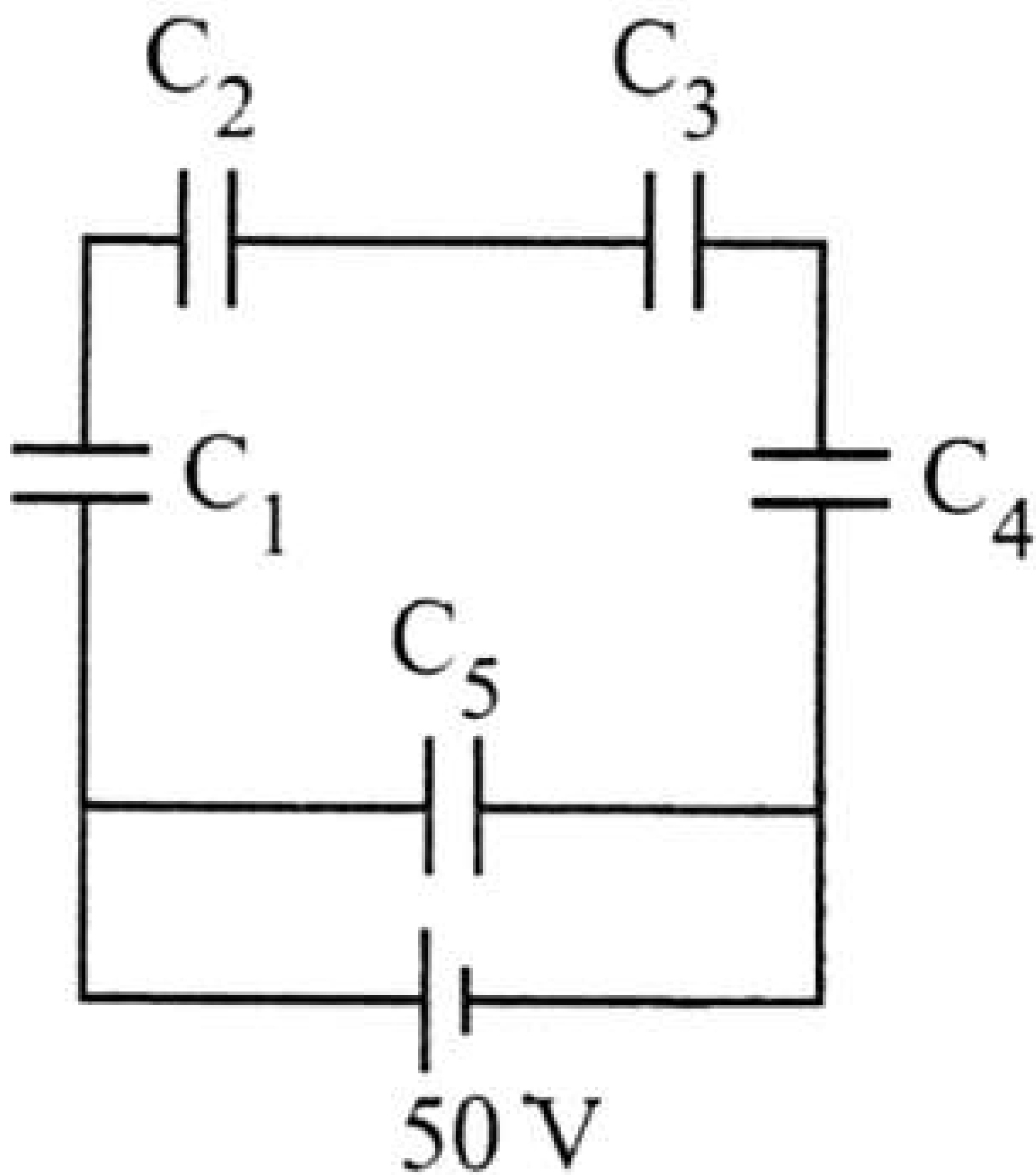












of Solid

Column-II

$$\frac{\Delta d}{d} \left(\frac{L}{\lambda} \right)$$

4) $5 \mu\text{F}$, $125 \mu\text{F}$

Sol. (4):

10. The amount of work done in pulling the chain from the surface to the radius

$$1) \quad mg \frac{R}{2}$$

of Solid

column-II

$$\Delta d \left(\frac{L}{\gamma} \right)$$

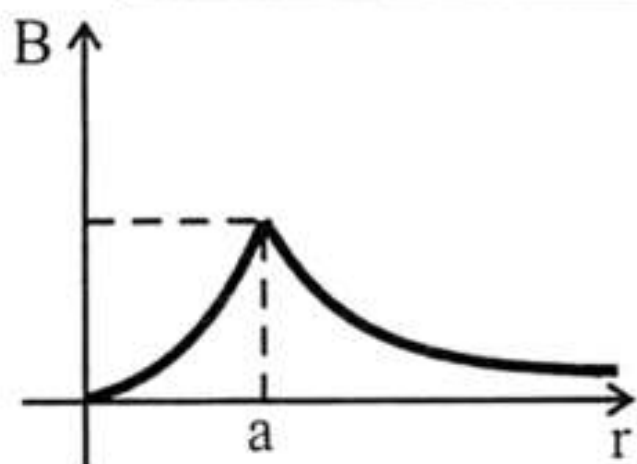
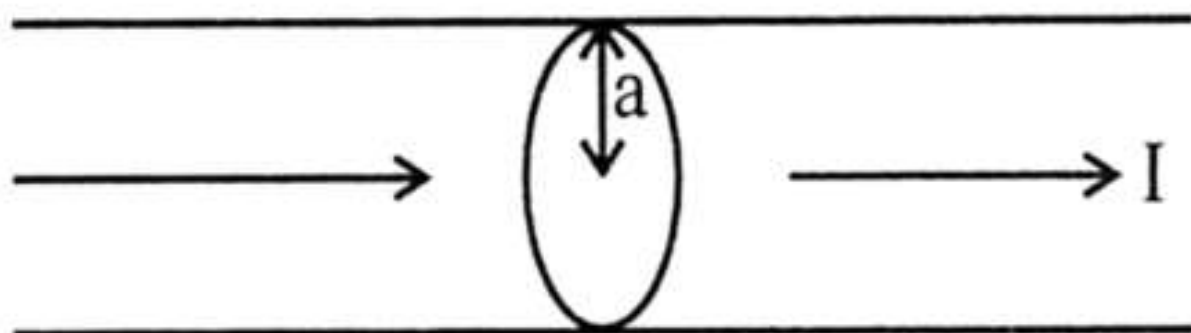
C** RCC** RCC** RCC** RCC

$$4) 5 \mu\text{F}, 125 \mu$$

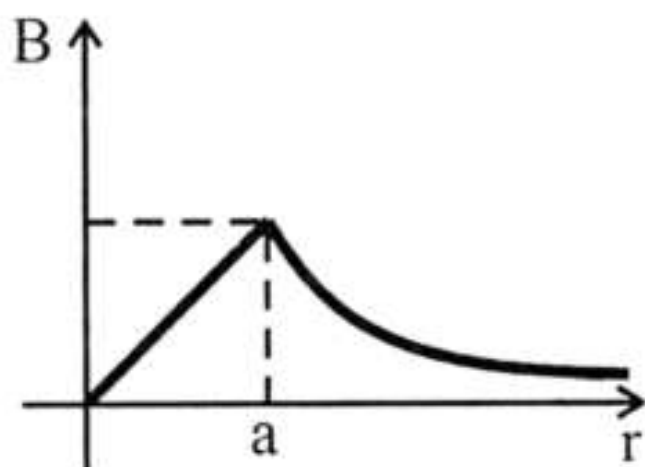
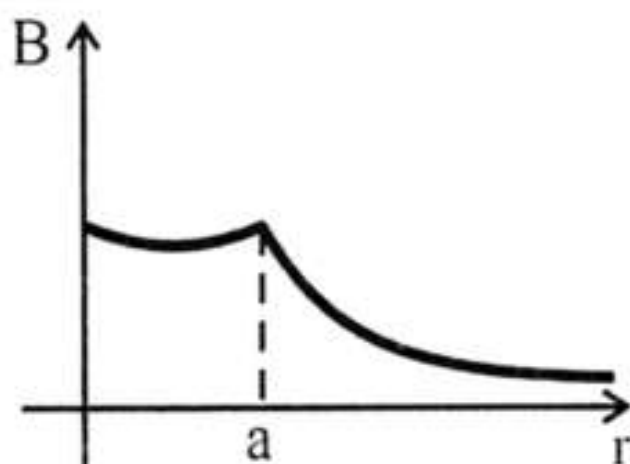
Sol. (4):

10. The amount
from the surt
to the radius

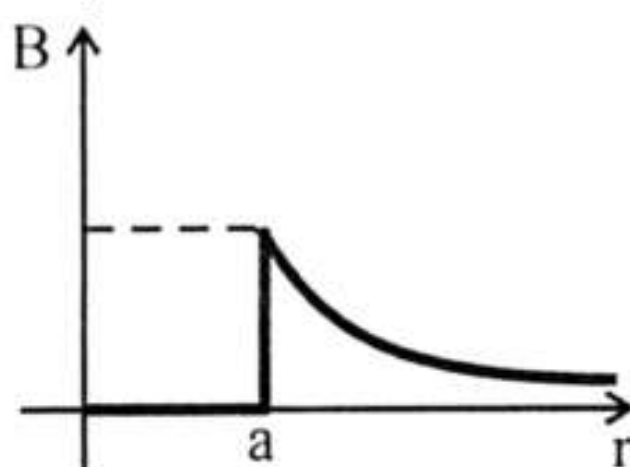
$$1) mg \frac{R}{\gamma}$$



2)



4)



Resistance of $1\text{ k}\Omega$, a
inductor of 1 mH
resonance frequency
is:

7 kHz

13.5 kHz

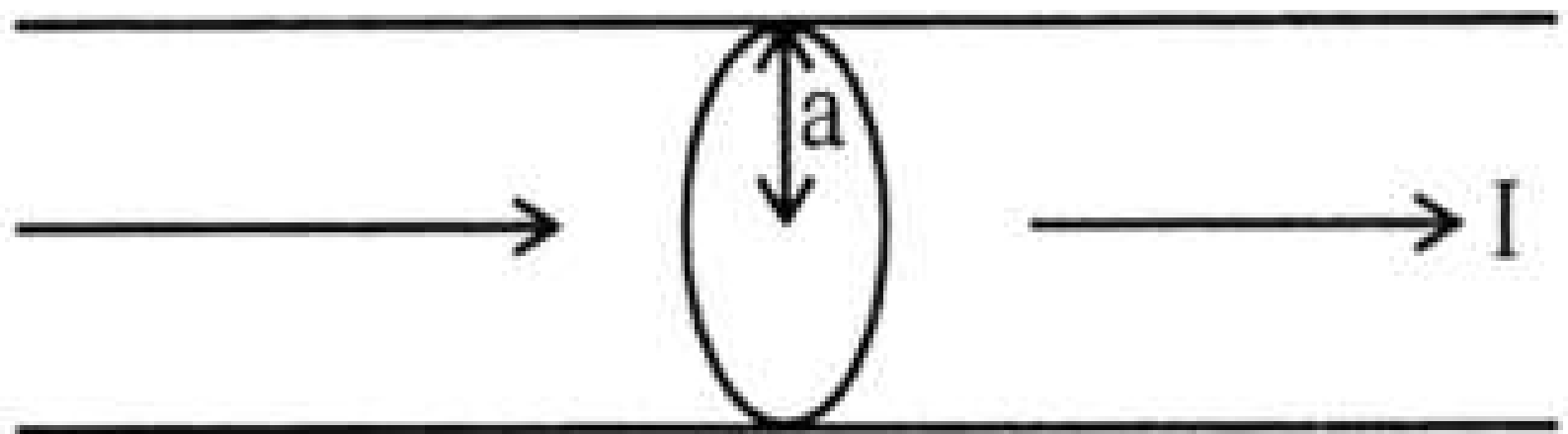
Electric Current

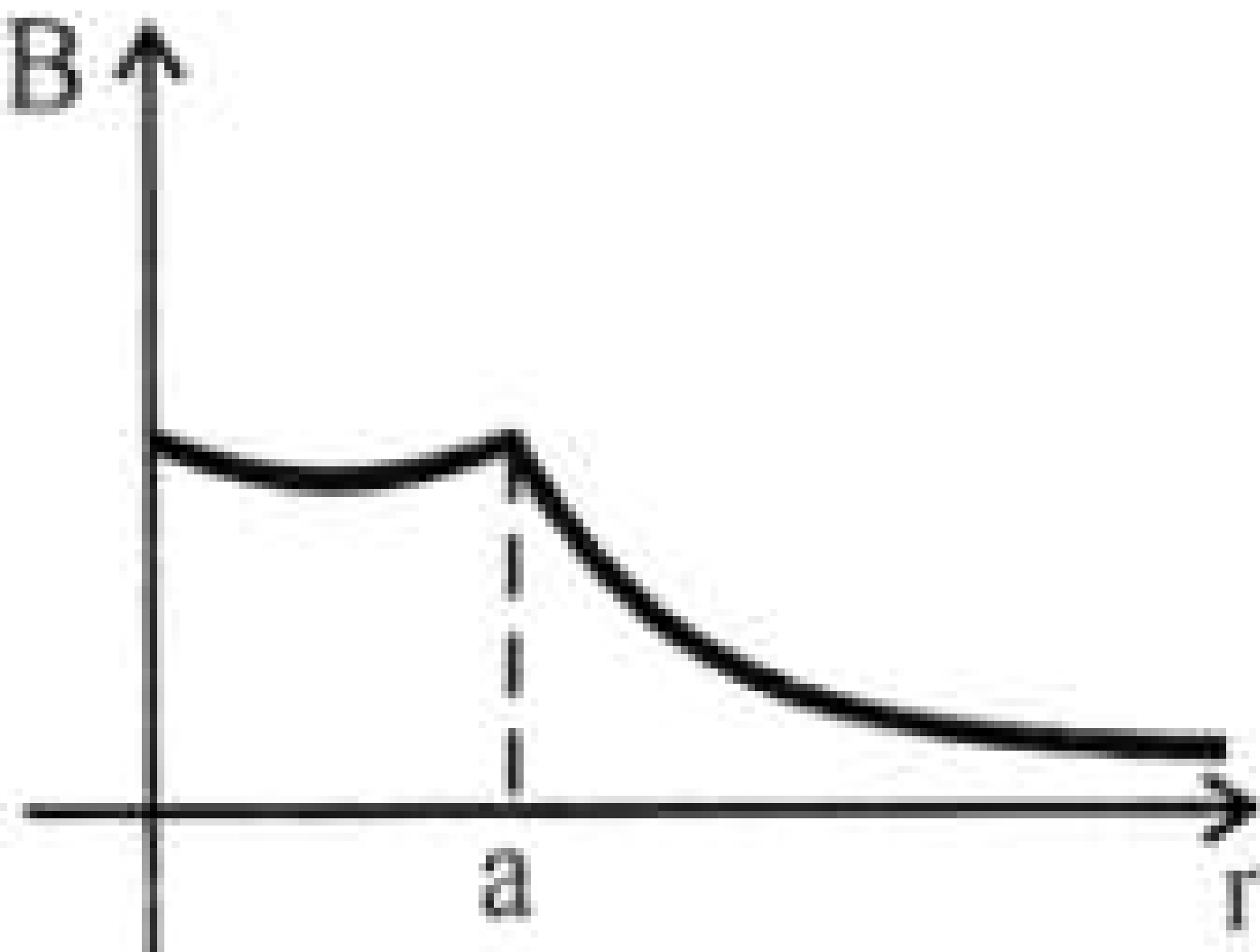
shows a long straight

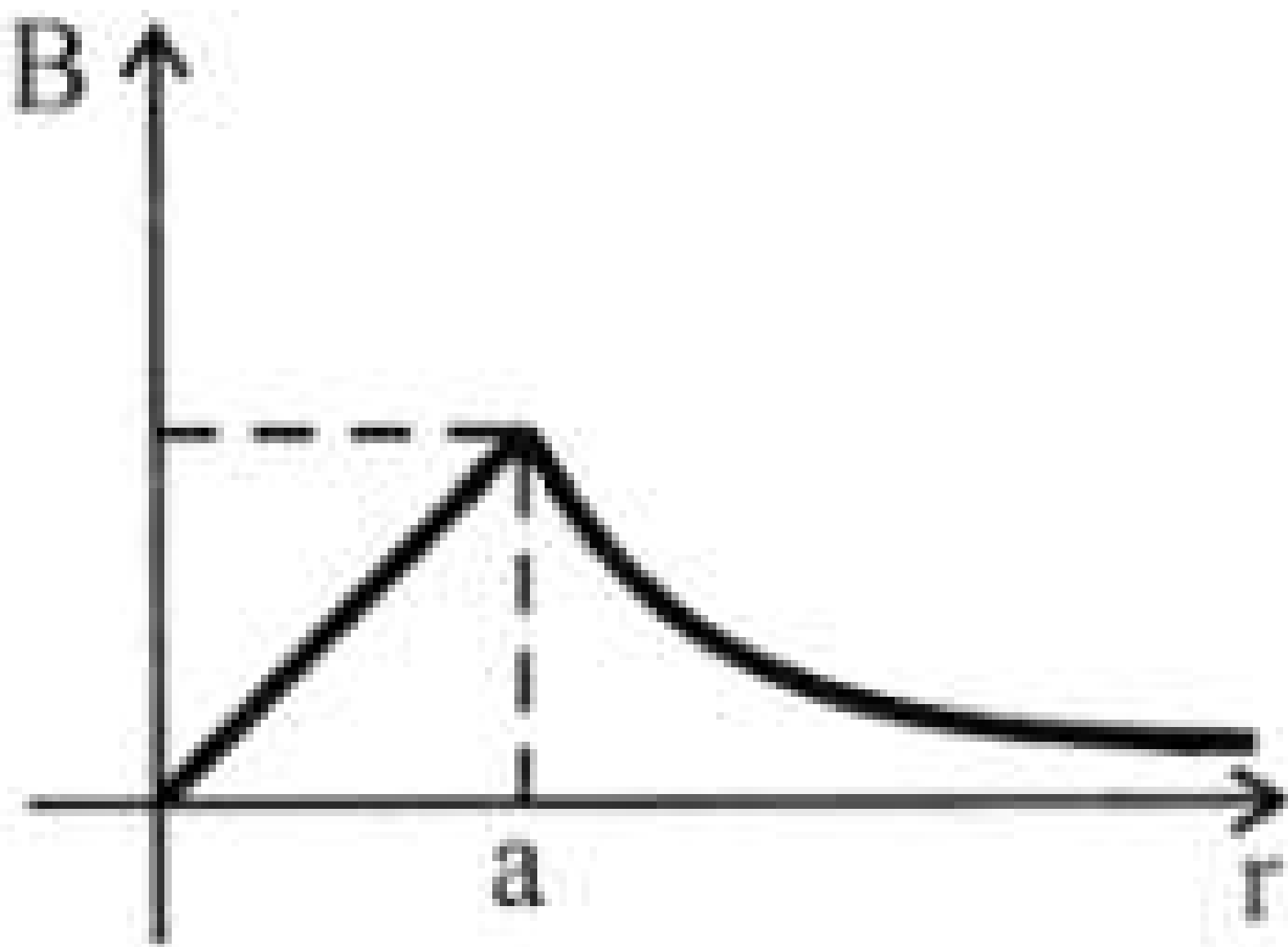
RCC** RCC** RCC** RCC** RCC**

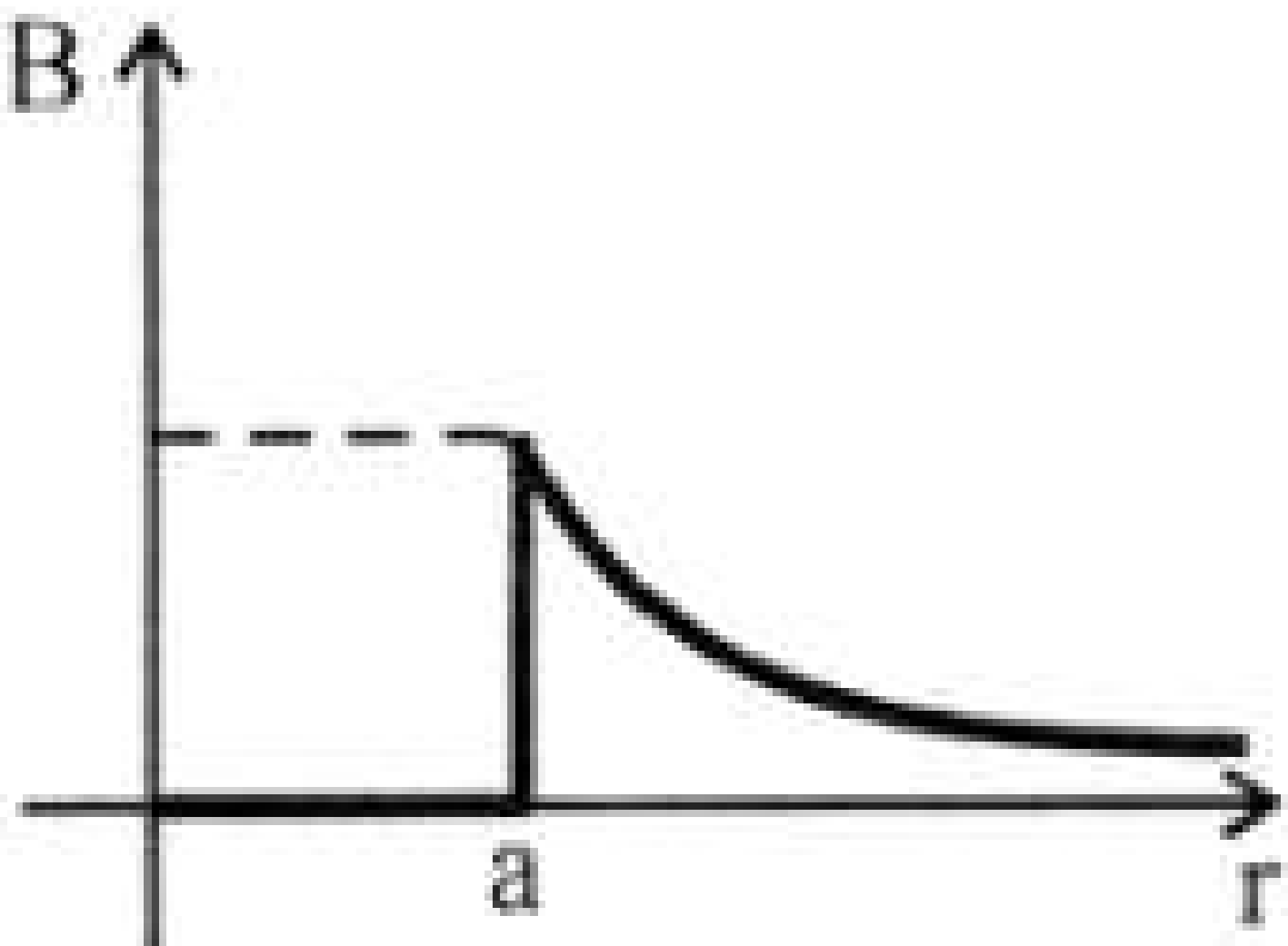
18. In Young's double-slit experiment, the intensity of the central maximum is I_0 . The intensity of the n th minimum is I_n . The ratio I_n/I_0 is

1) K









a battery of 12 V
2 Ω . If the current
terminal voltage of

V
8 V

ty

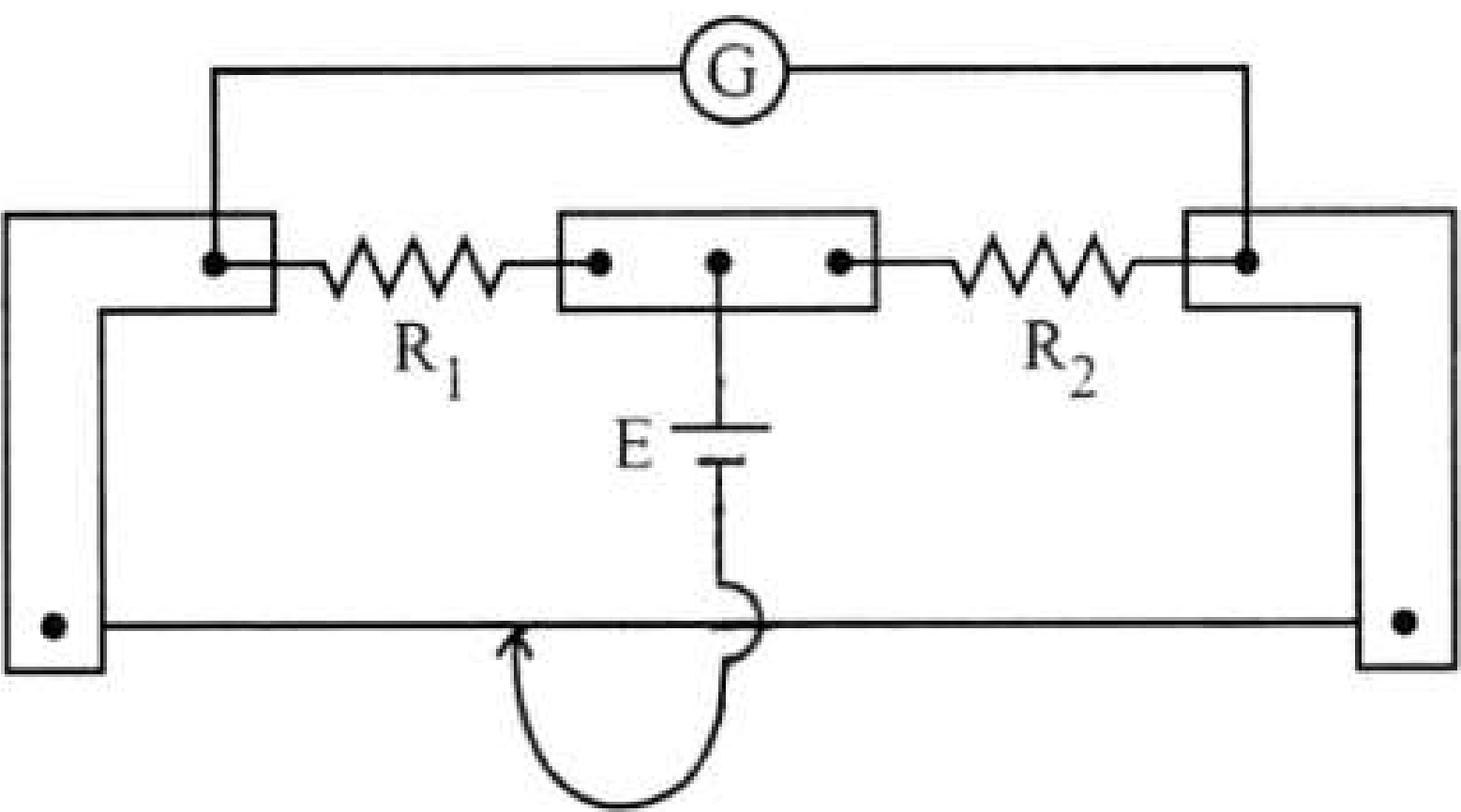
RCC** RCC** RCC** RCC** RCC**

C. The metal
have no
situation.

D. At the sur
electrostat
surface at

E. The elec
everywhere

Choose the c
given below



battery of 12 V
 Ω . If the current
 terminal voltage of

V

V

y

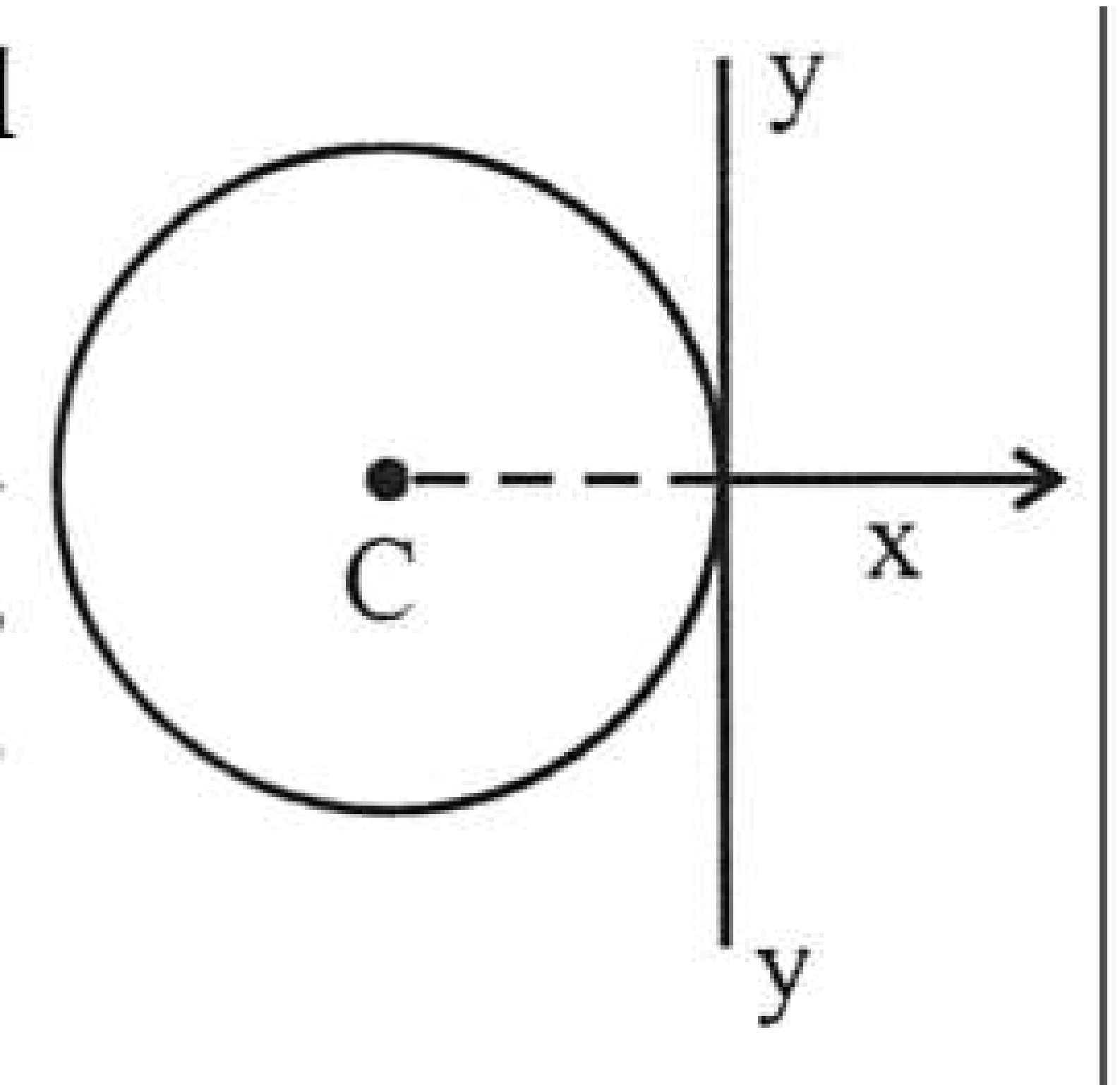
C** RCC** RCC** RCC** RCC

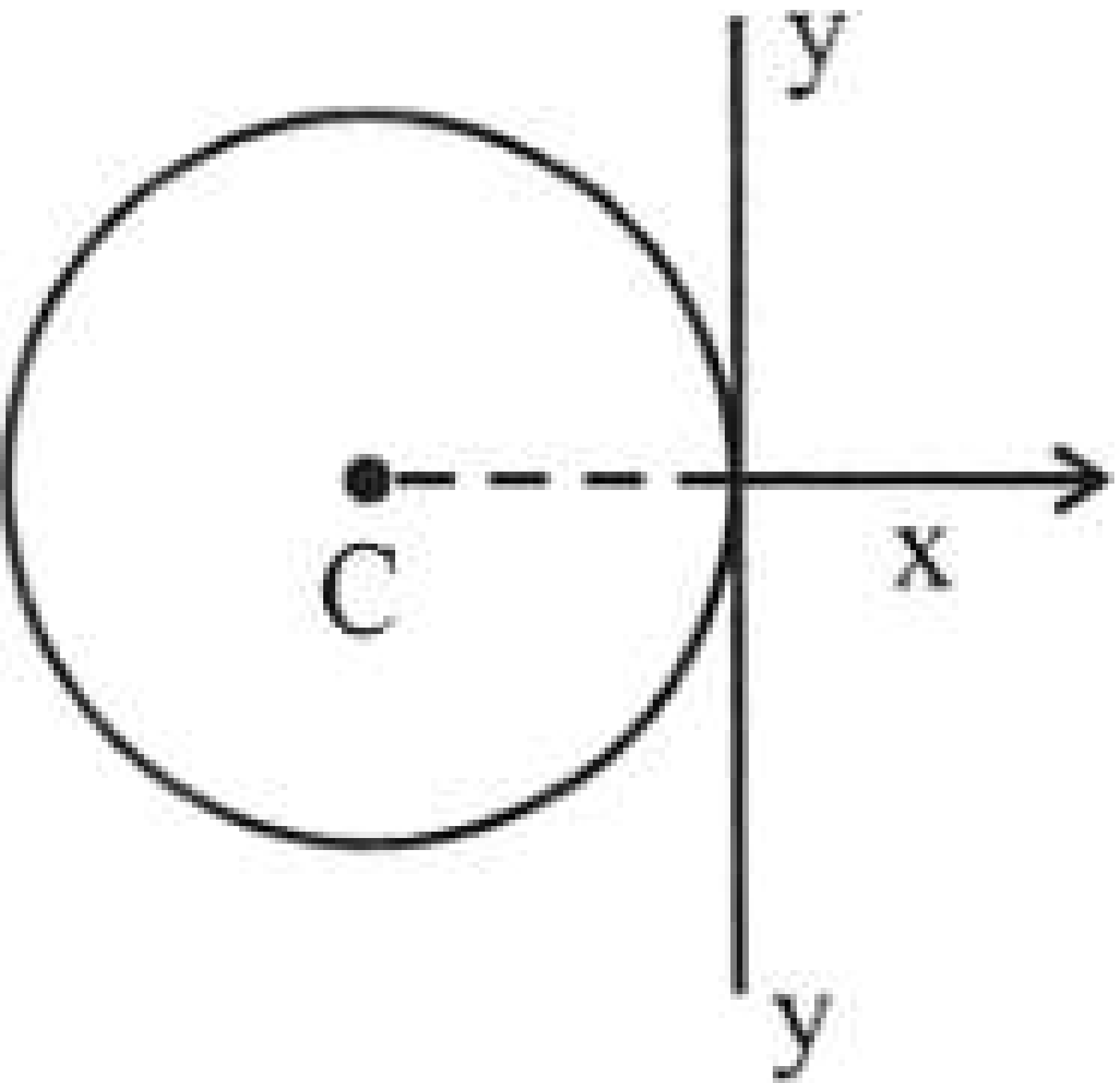
have no
 situation.

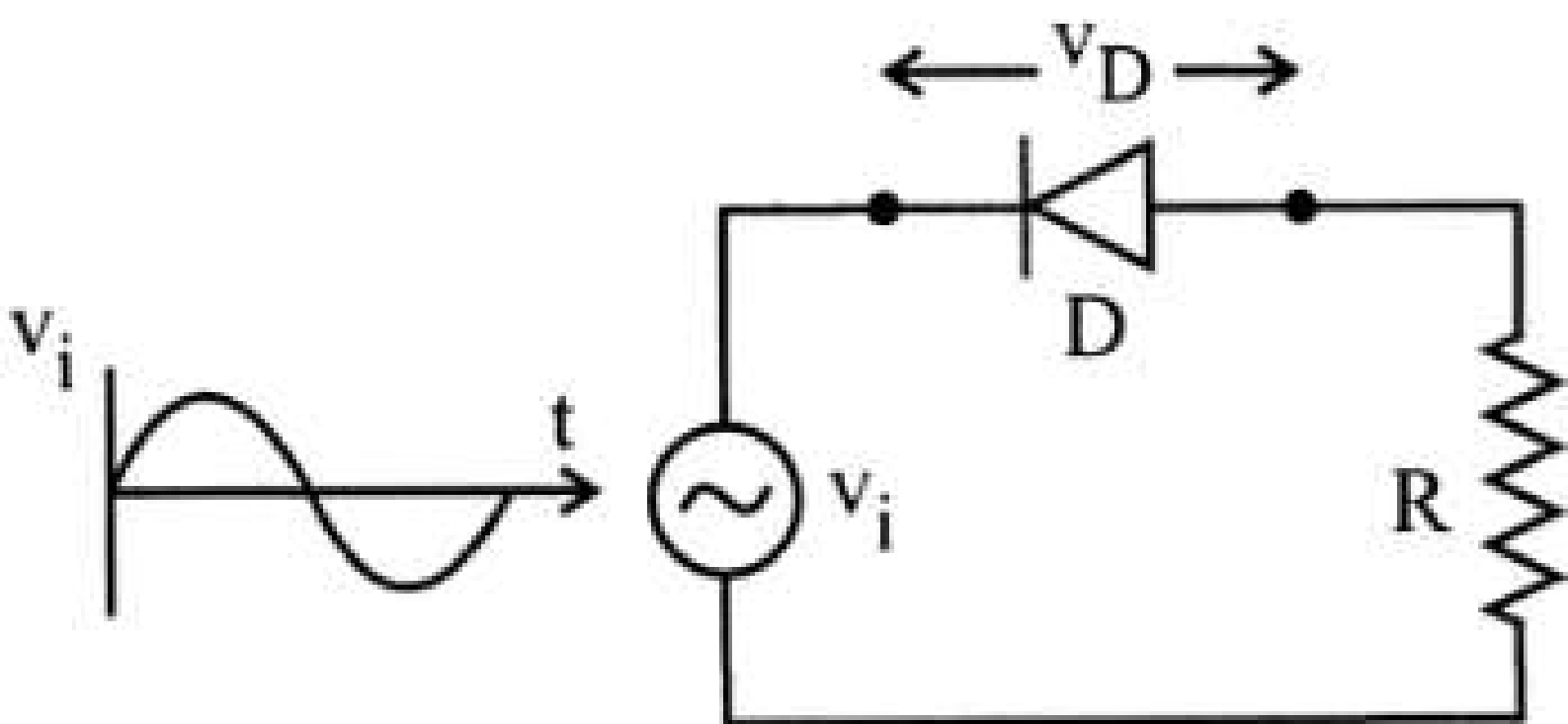
D. At the sur
 electrosta
 surface at

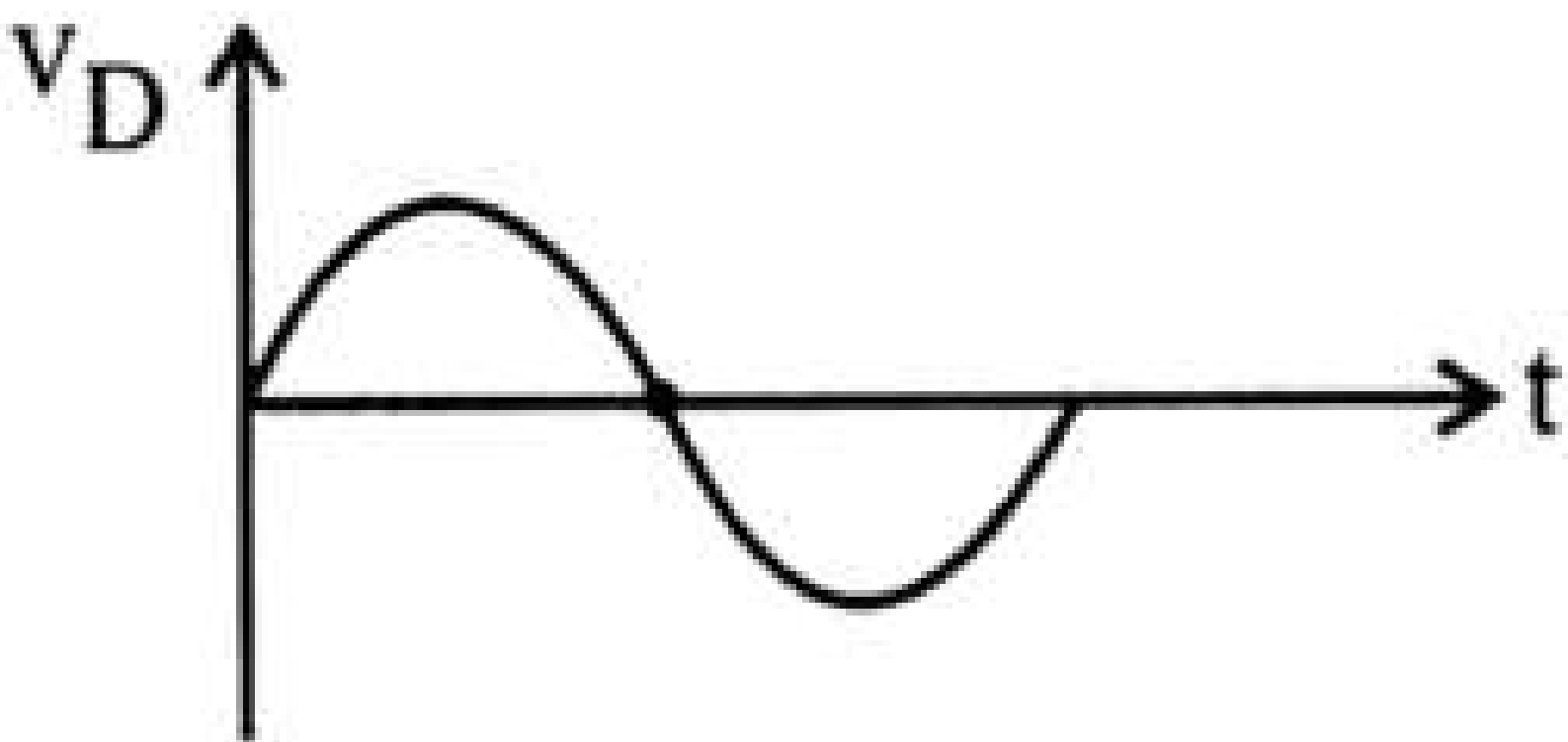
E. The ele
 everywhe

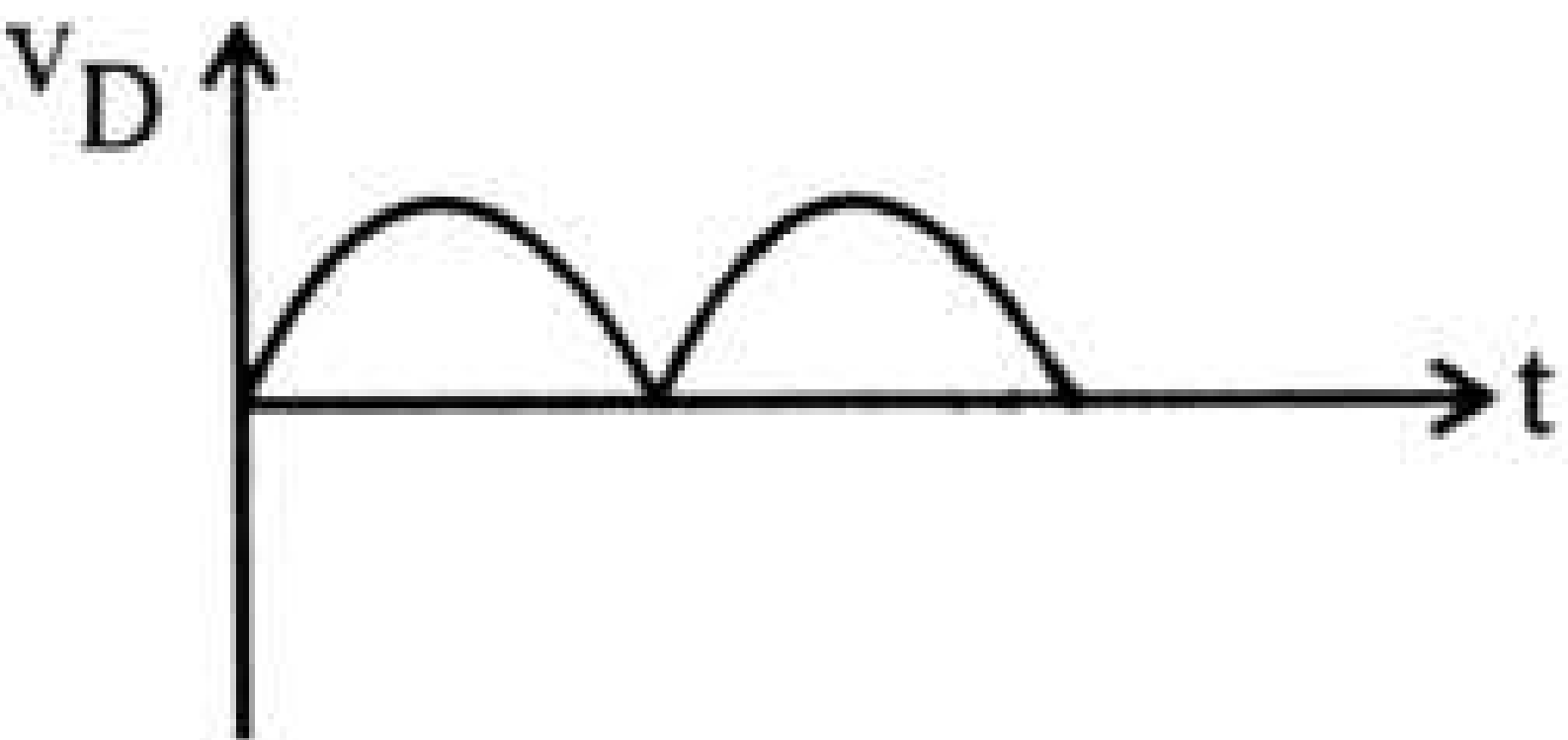
Choose the
 given below

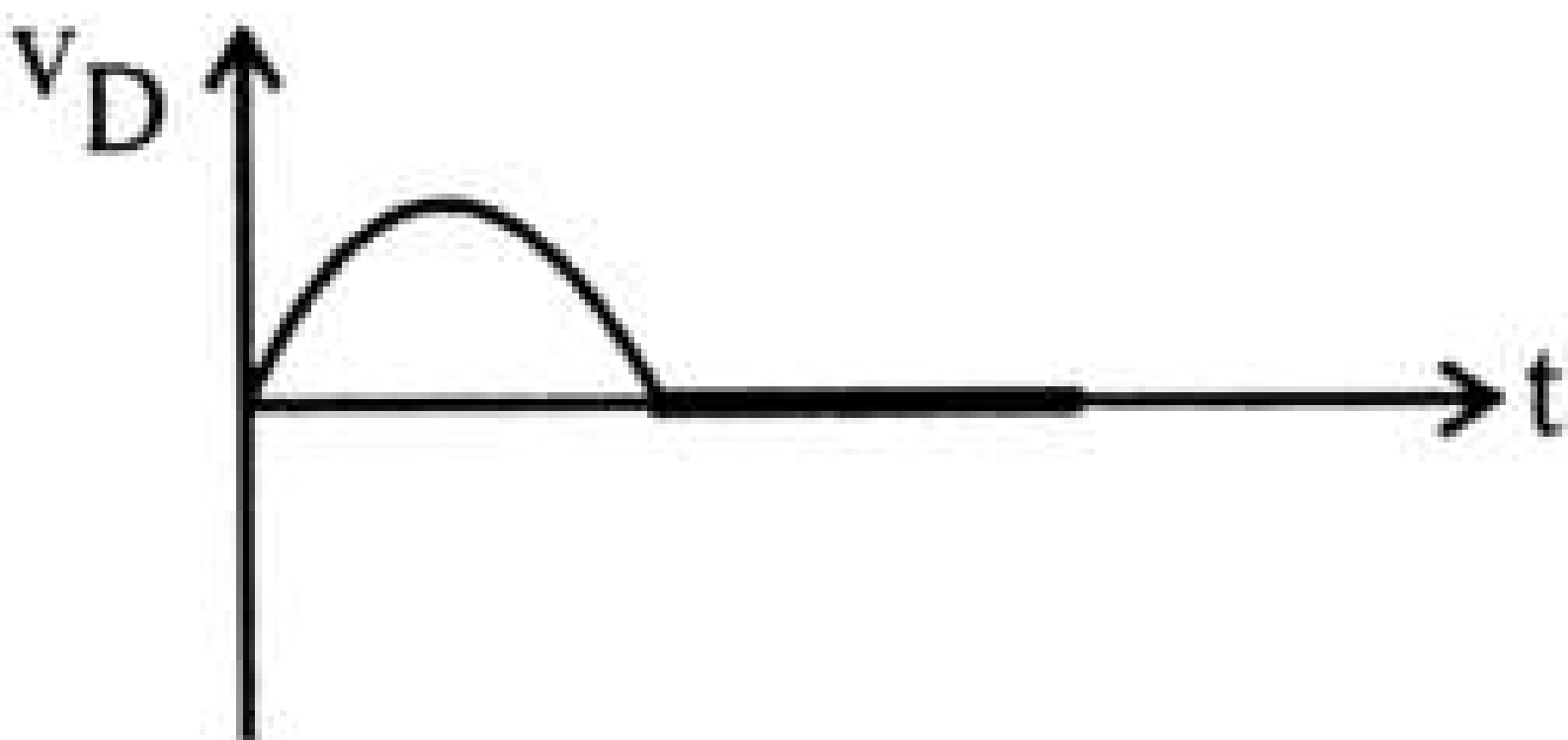












c Current

$100\ \Omega$ gives full
of 1 mA . It is
of range $0 - 10\text{ A}$.

Ω

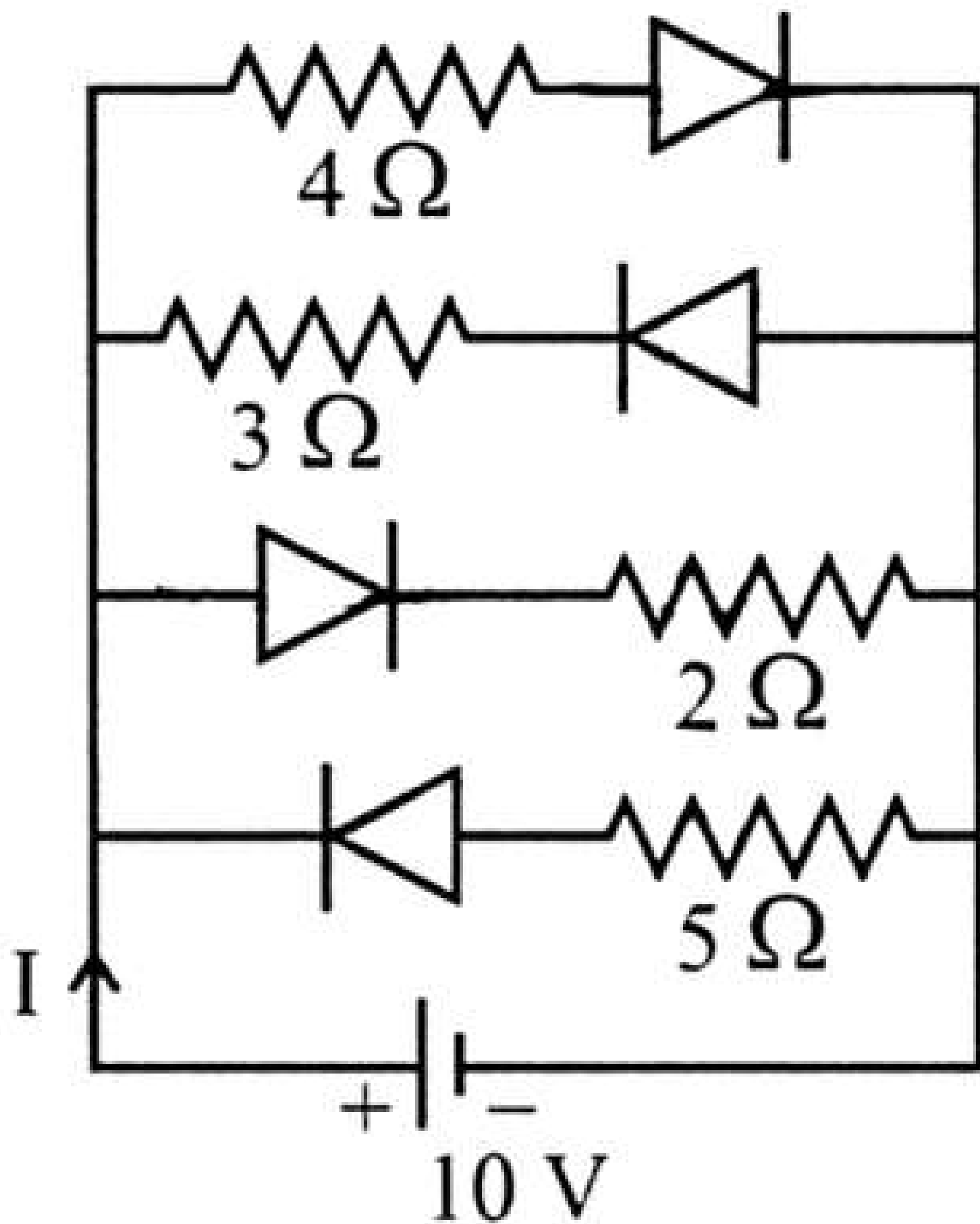
C** RCC** RCC** RCC** RCC** RCC

Sol. (2):

Sem

32. In the circuit
appearing ac
form:

v_i



in atoms emit
when they move
higher energy
lower energy

ve decay of

a of atoms and

RCC** RCC** RCC** RCC** RCC**

Dual Nature

37. For a metal of
the following
does not give
(Take Planck's

1) 200nm

3) 50 nm

Sol. (1):

Unit

in atoms emit
 when they move
 higher energy
 lower energy

ve decay of

of atoms and

C** RCC** RCC** RCC** RCC

Dual Nature

37. For a metal of work function ϕ , the following wavelengths of light does not give photoelectric emission. (Take Planck's constant $h = 6.6 \times 10^{-34} \text{ Js}$)

- 1) 200nm
- 2) 400nm
- 3) 50 nm
- 4) 100 nm

Sol. (1):

Unit

and E only
only

RCC** RCC** RCC** RCC** RCC**

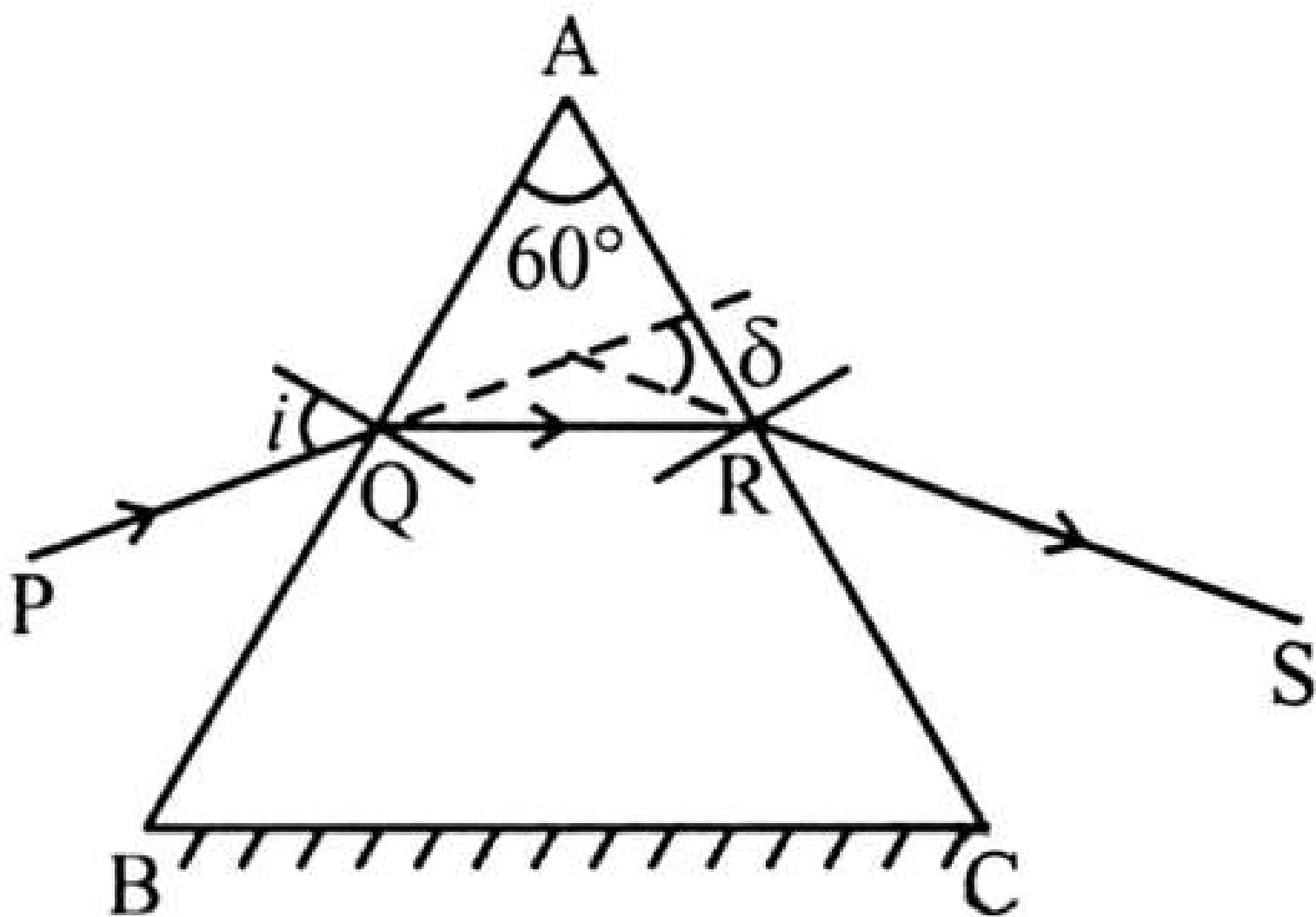
energy of its
distance of t
nucleus in th

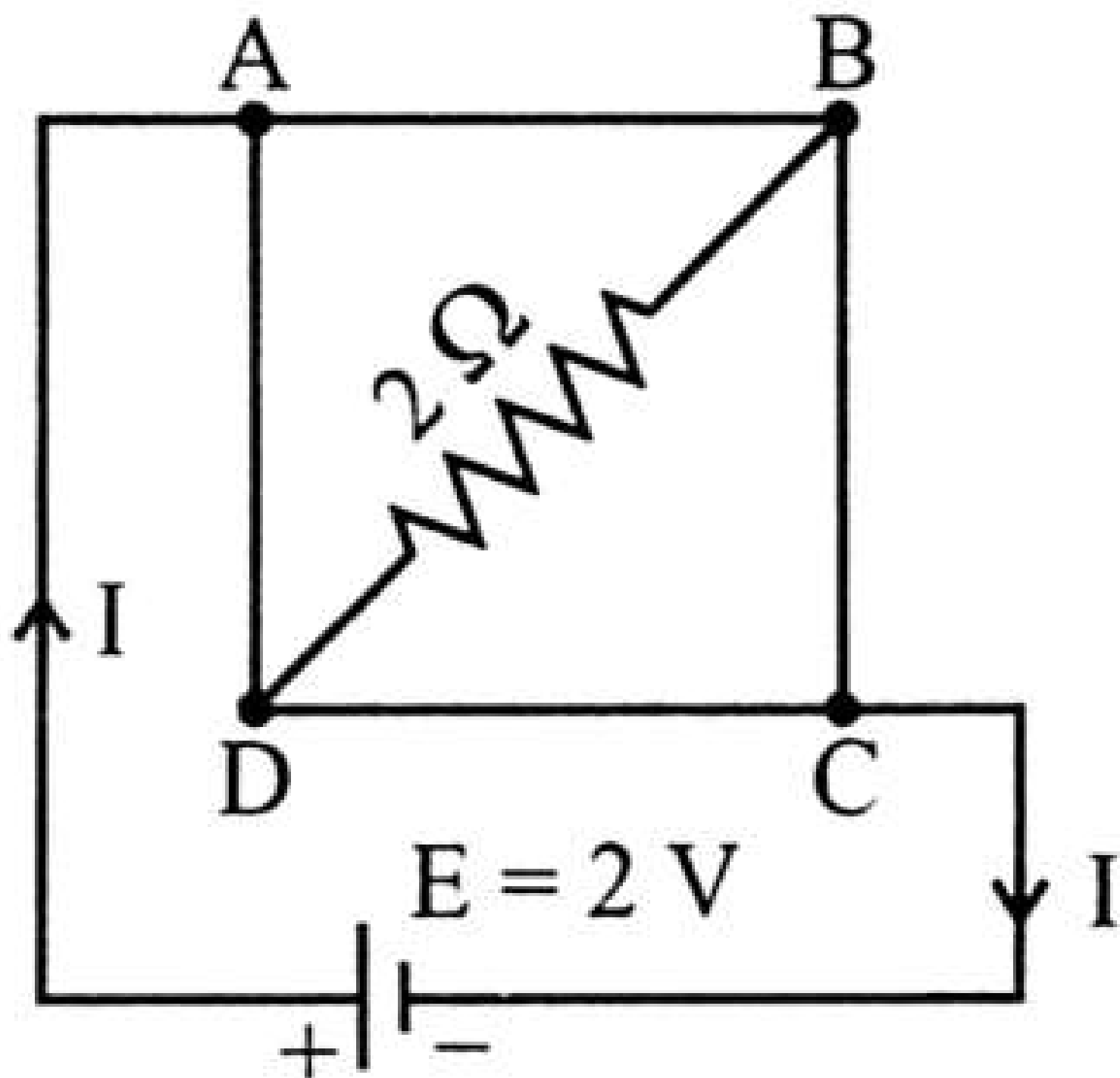
(Take 1 eV =

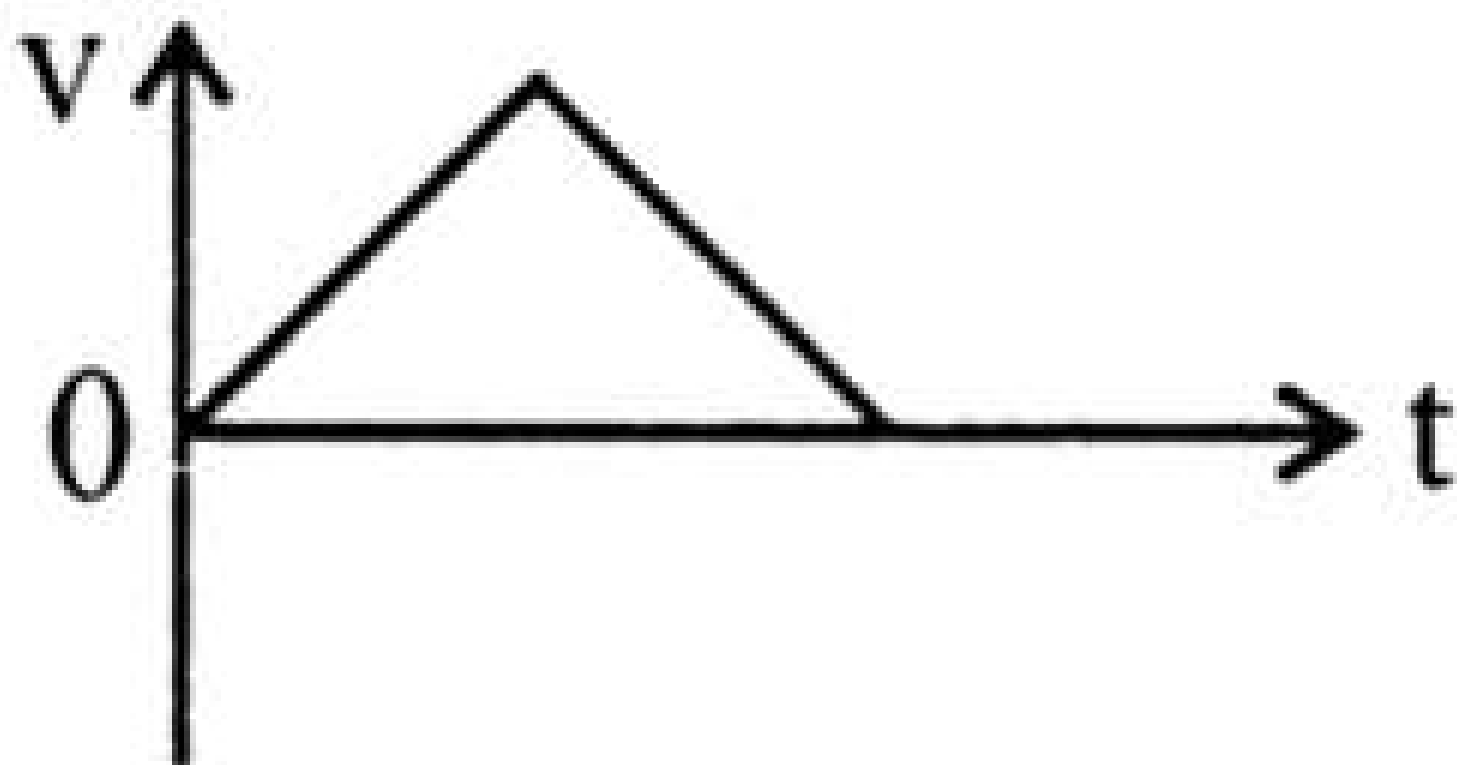
$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

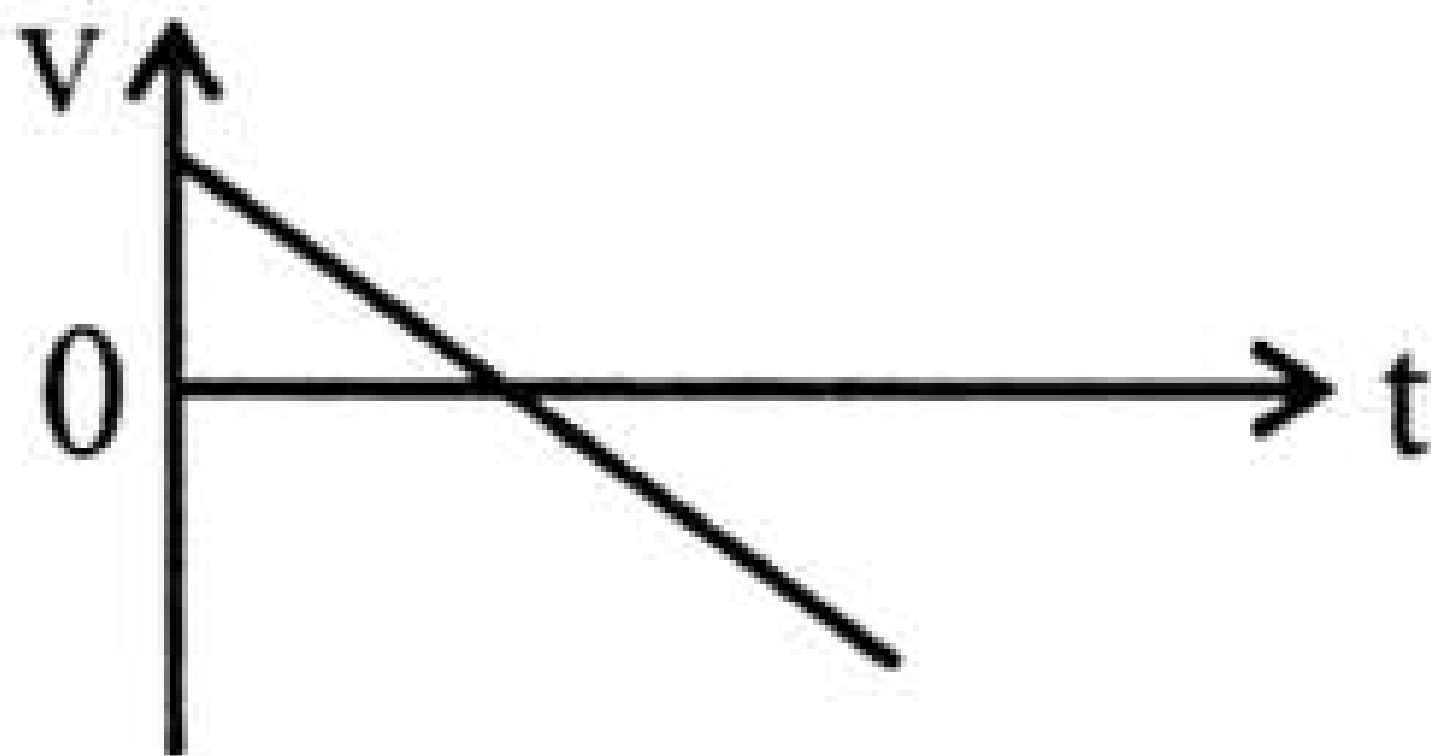
1) 2.1×10^{-8} m

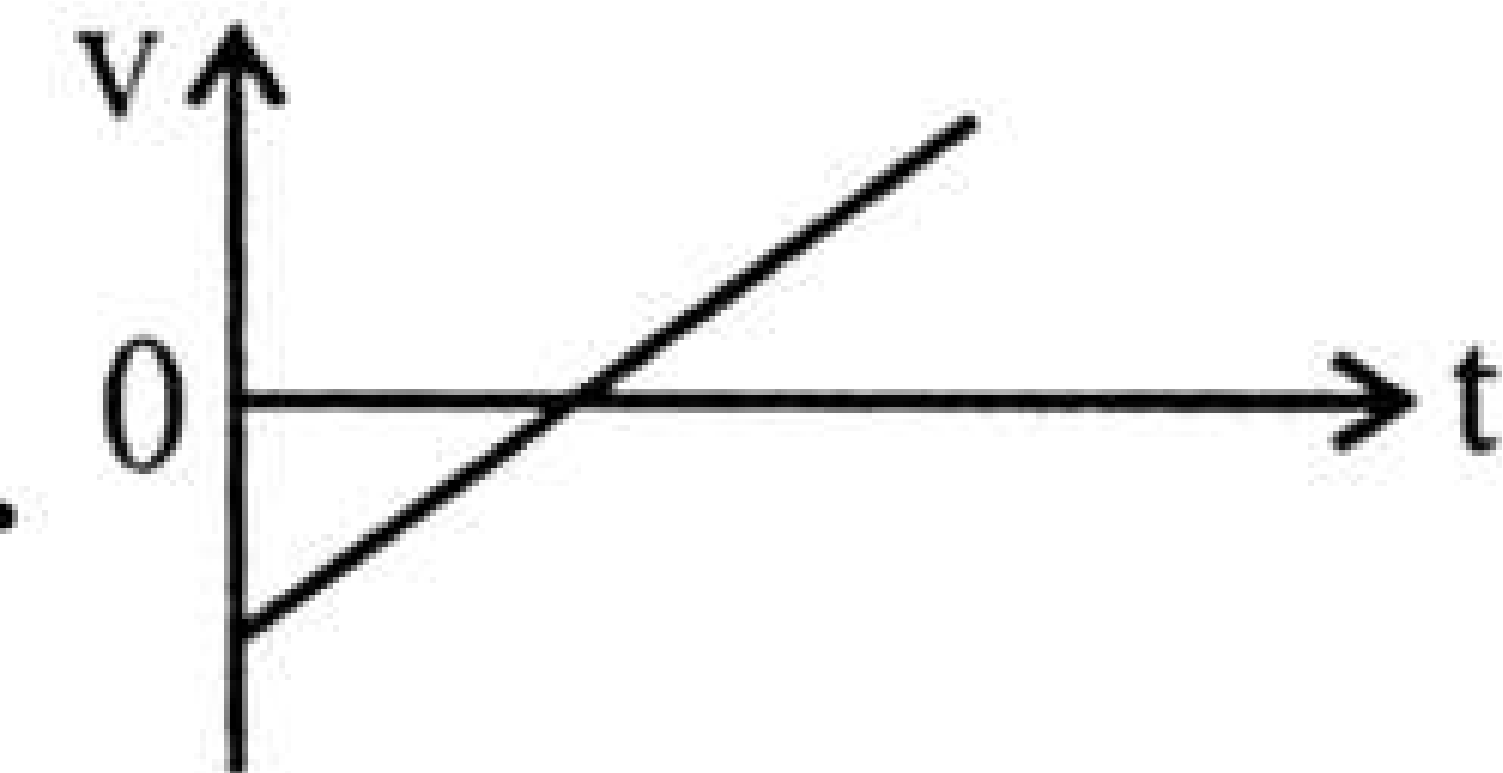
3) 2.1×10^{-11}

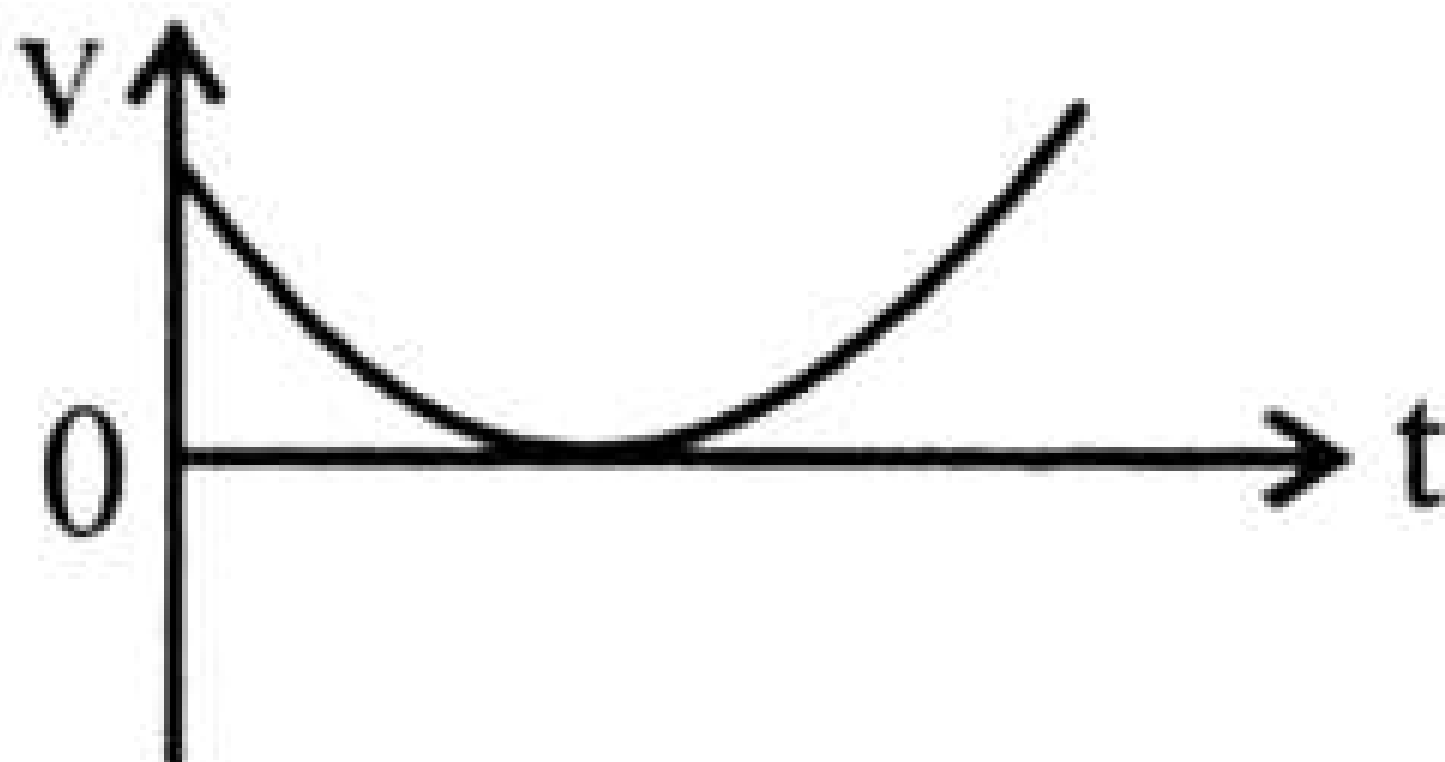


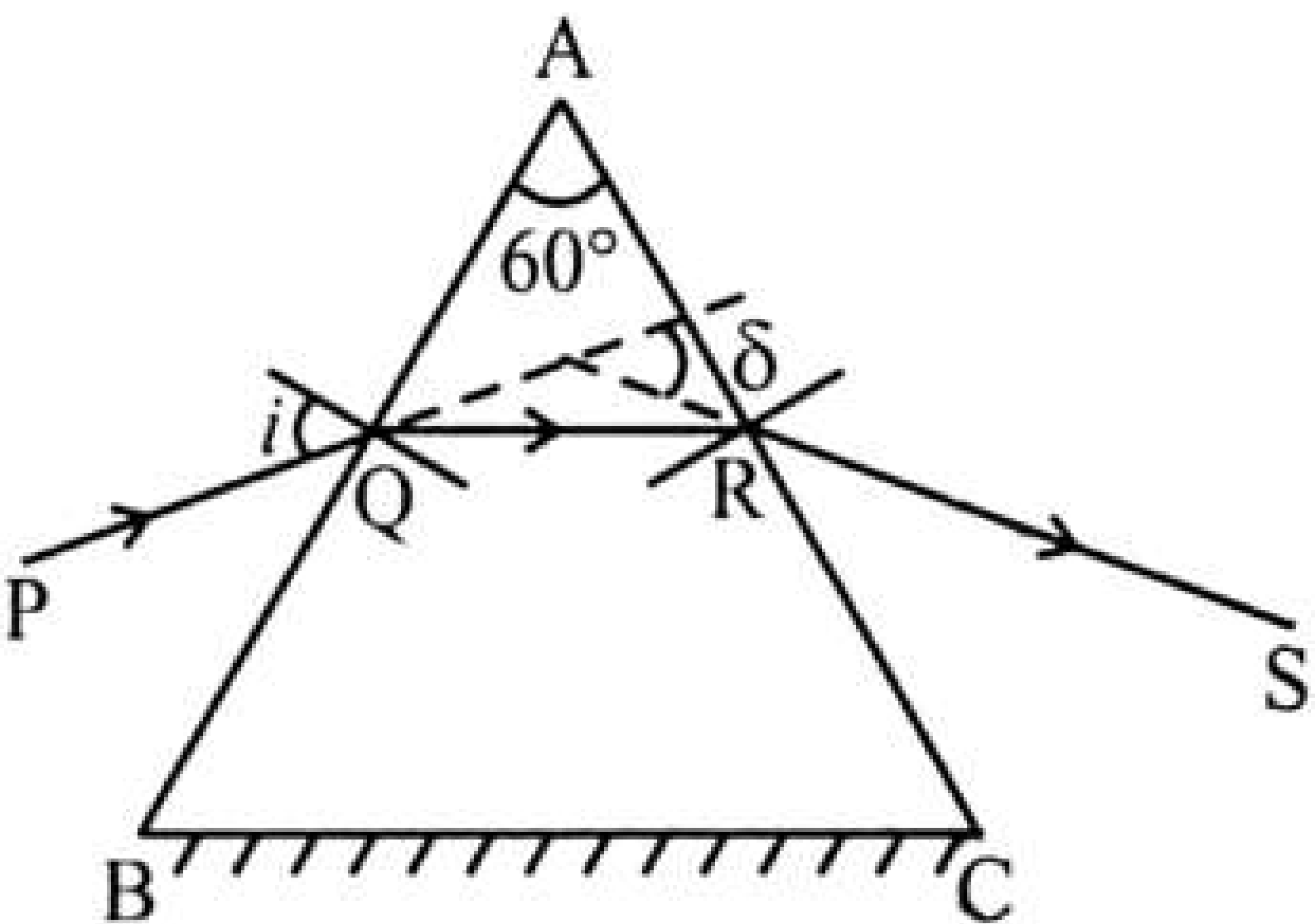


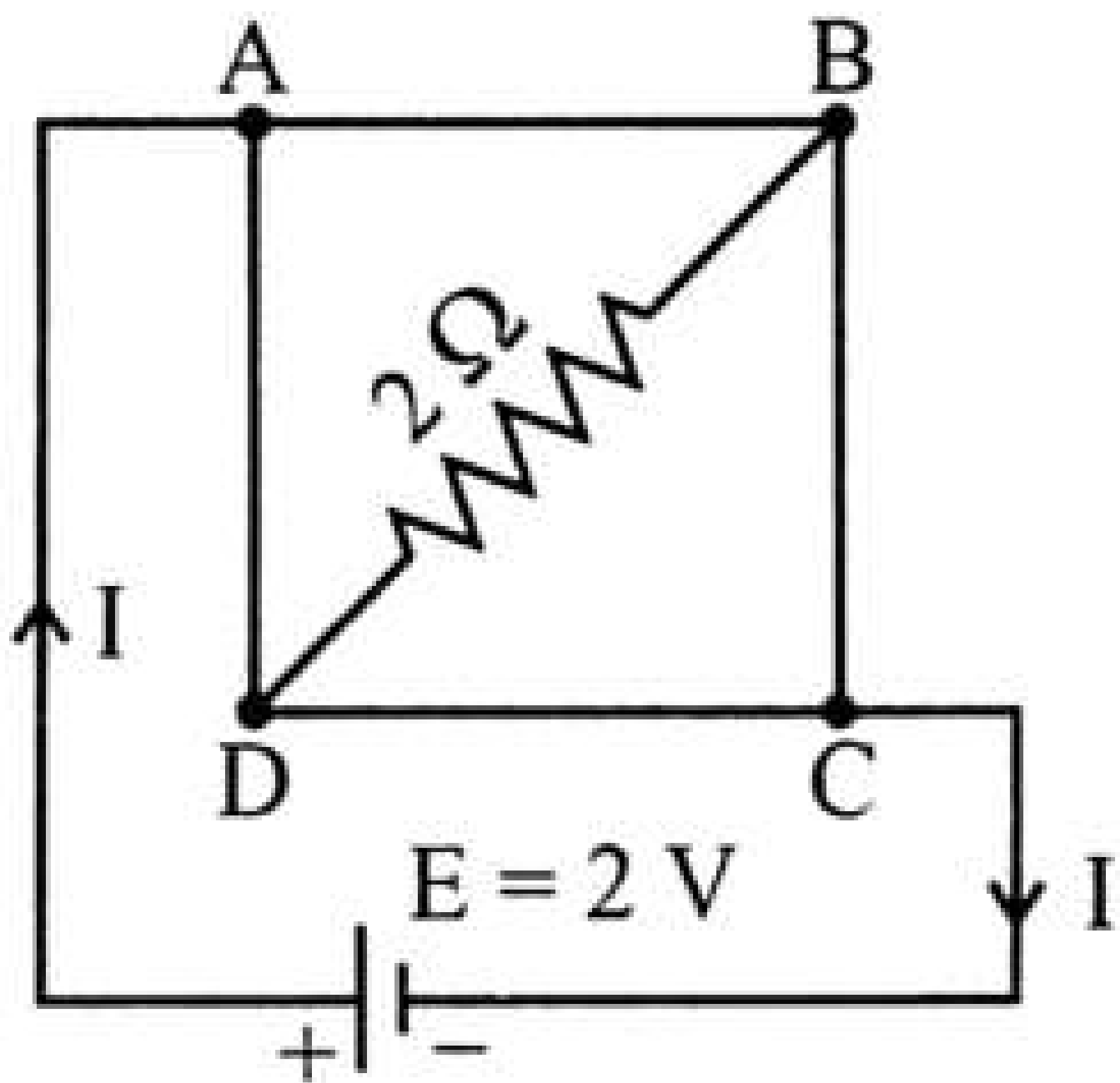


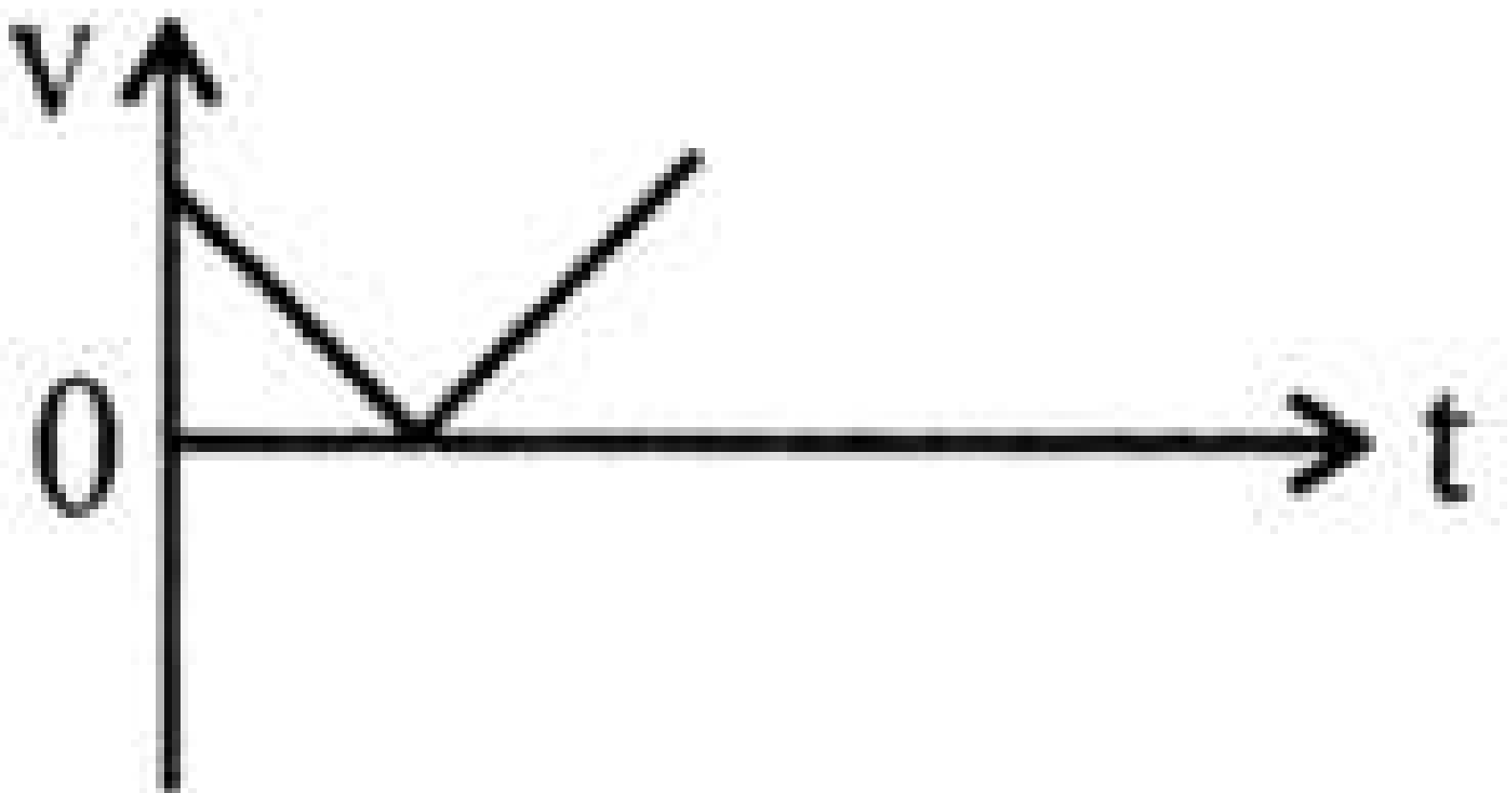


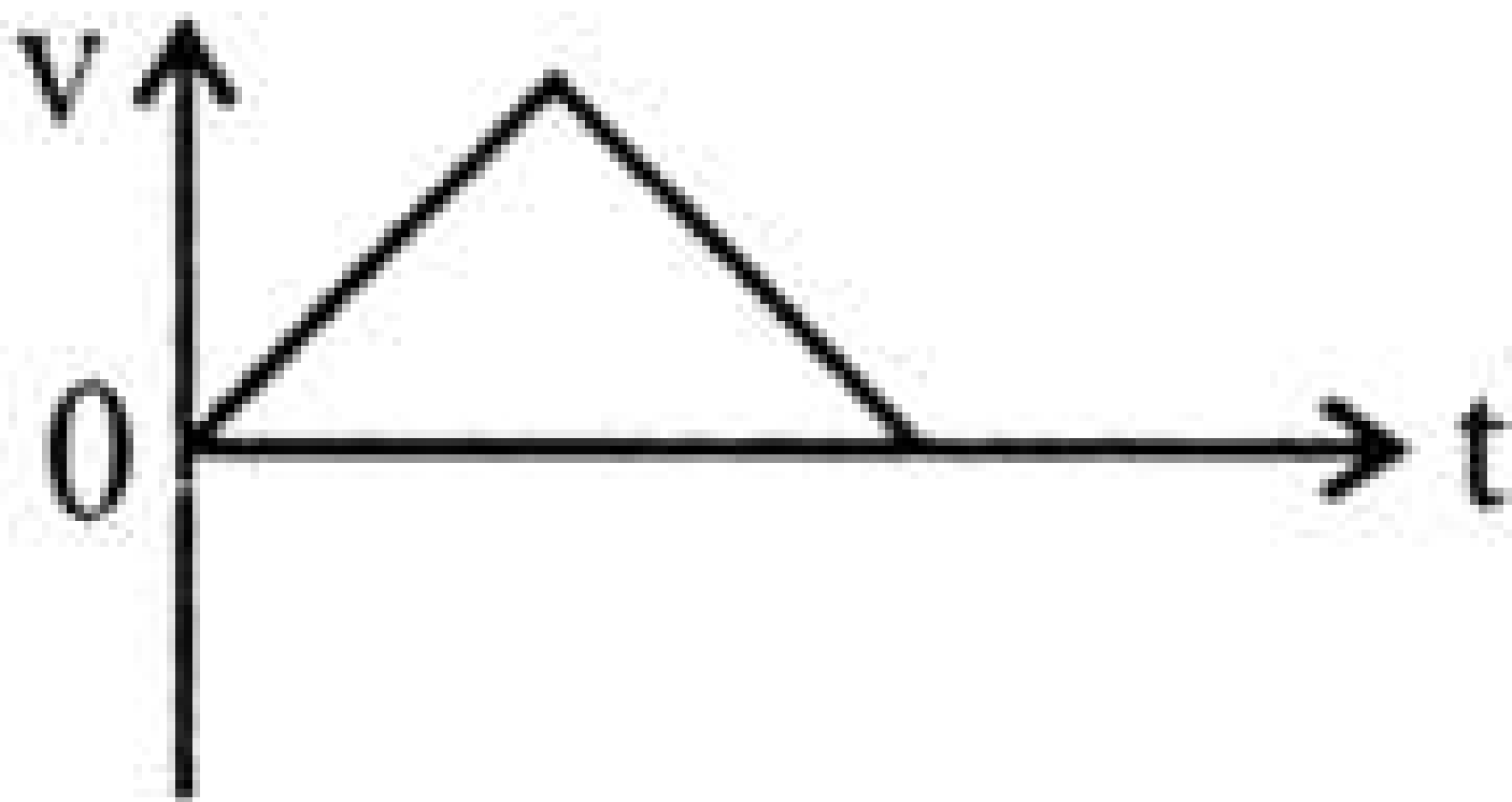


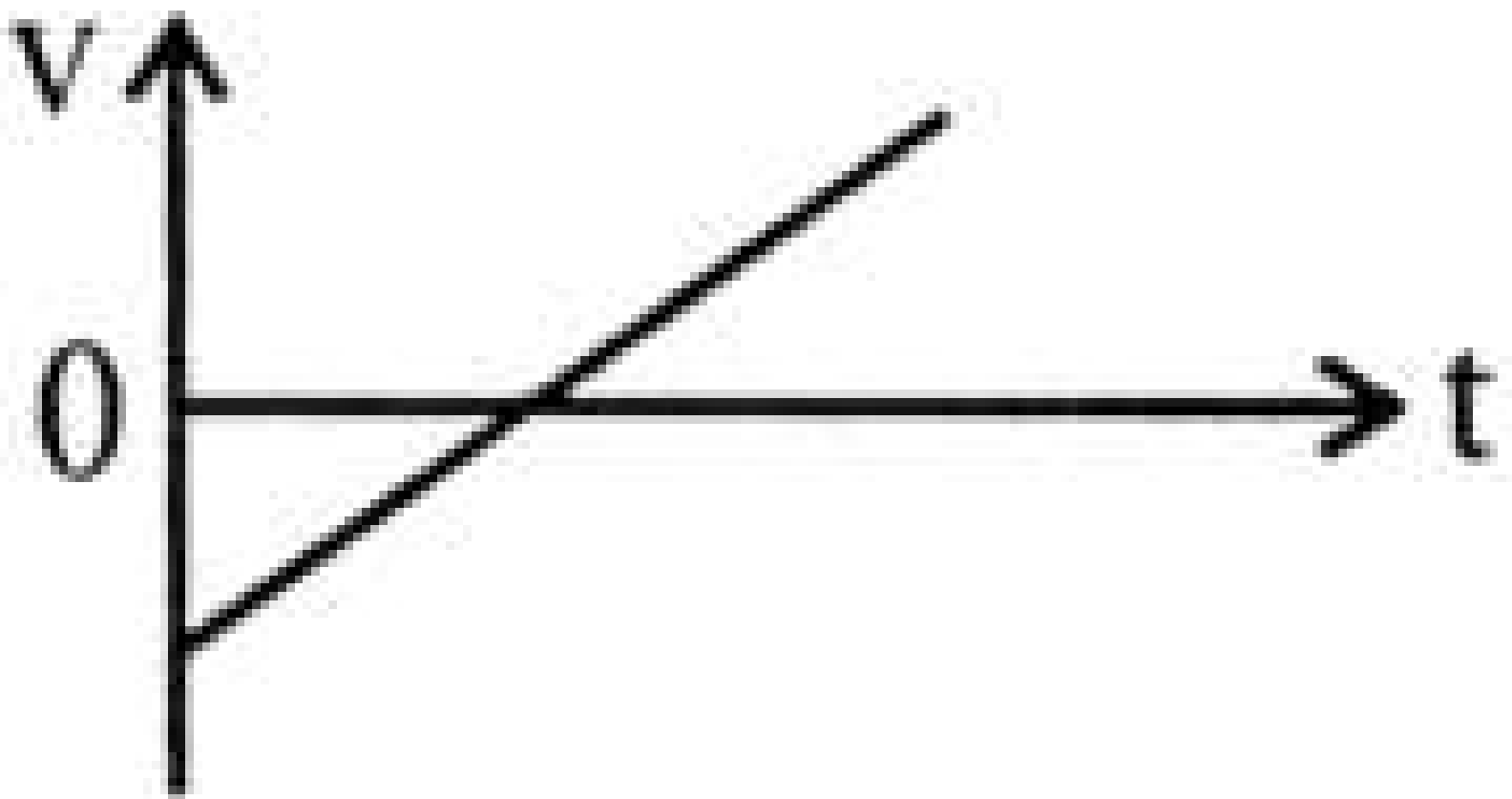


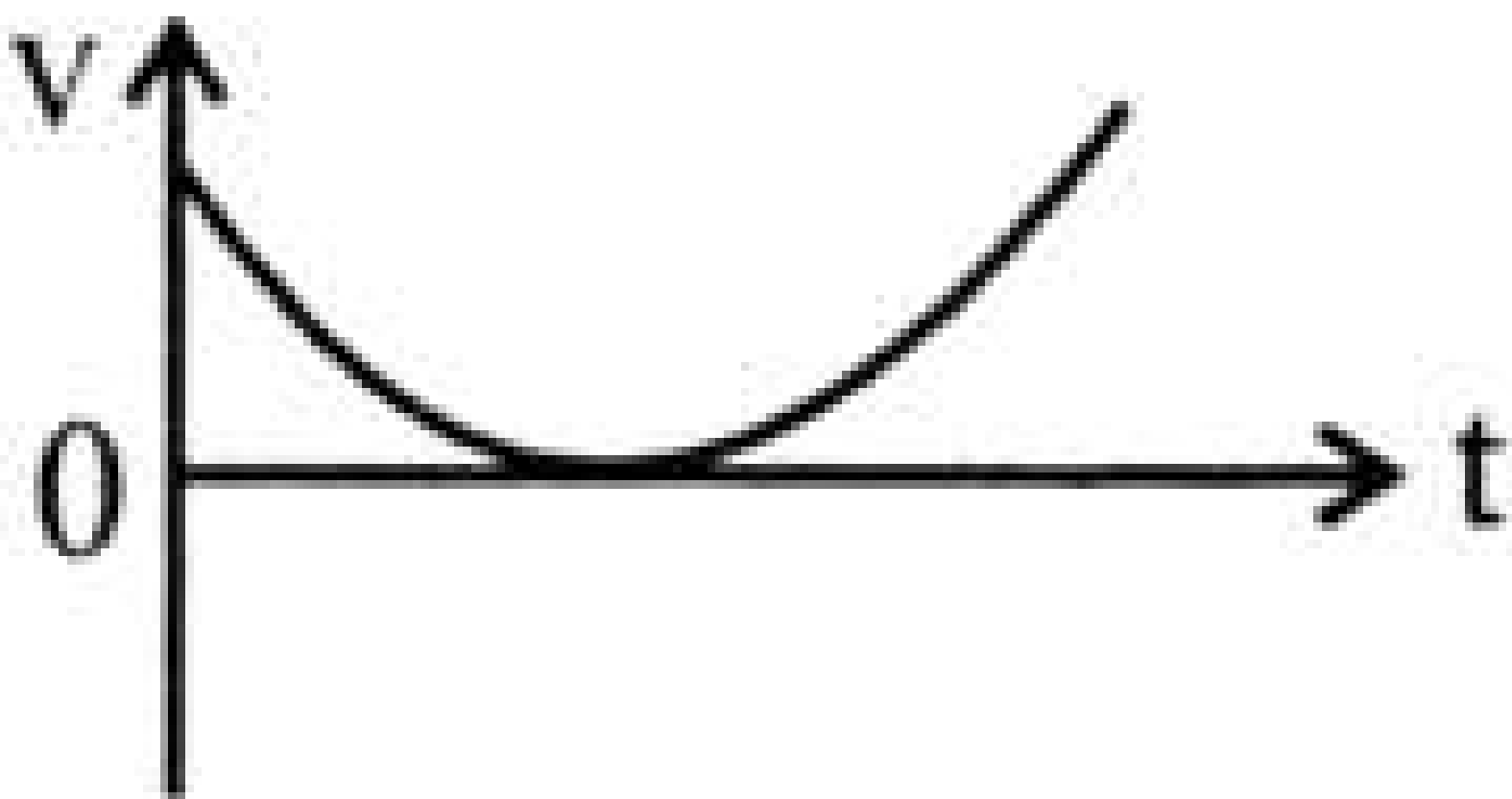


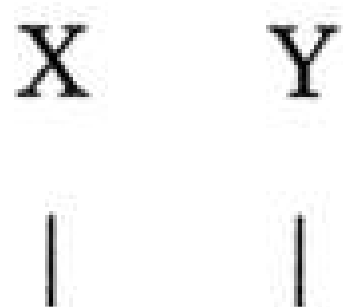












(Substrate)

(Product)

(Product)

